Geophys. J. Int. (2026) **000**, 1–21

An analytical recurrent-steady-state model for inter-rifting deformation at magmatic rifts

Iván Navarrete[1]
Jean-Arthur Olive[1]
[1] *Laboratoire de Géologie, École Normale Supérieure, PSL Research University, Paris, France*

SUMMARY
Inter-rifting deformation, the strain that accumulates between discrete magmatic episodes, is the least-constrained phase of the rifting cycle. We model it as the superposed relaxation of periodic dike intrusions in an elastic plate overlying a Newtonian viscous half-space. The non-local coupling of the plate to the half-space turns post-diking relaxation into a fractional diffusion governed by the half-Laplacian. A single characteristic velocity, $a = EH/[2\eta(1-\nu^2)]$, controls this diffusion, where E, H and ν are the Young's modulus, thickness and Poisson's ratio of the plate and η is the viscosity of the half-space. Deformation spreads from the rift at the constant speed a with a $1/x$ tail. This solution is an average of Fickian diffusion profiles over a heavy-tailed distribution of diffusion times, of which the Fickian model retains only one. Periodic intrusions converge to a recurrent steady state (RSS) with a closed form in the Digamma and log-Gamma functions. Over each cycle the two flanks separate by exactly one dike opening, so the far field diverges at the long-term opening rate without lateral boundary conditions, and the deformation half-width aT, where T is the recurrence interval, is the only length scale. A Maxwell visco-elastic half-space with the shear modulus of the plate modifies this response only within about $1.3\,H$ of the axis and during the first few relaxation times. We fit post-rifting records from Krafla, Iceland, and Asal, Djibouti, deriving T from the fitted dike opening and a long-term rift opening rate. At both sites the viscous half-space fits better than the visco-elastic one, strongly so at Asal (ΔAIC $= 70$), whose record starts early in the relaxation. For $H = 10$ km and $E = 40$ GPa, the fits imply $\eta \approx 4 \times 10^{18}$ Pa s at Krafla and 2×10^{19} Pa s at Asal, with an uncertainty dominated by E. At Krafla, the RSS also fits better than Fickian diffusion ($\chi^2_r = 1.6$ against 4.9). Irregular recurrence, openings and dike positions spread individual velocity profiles around the RSS with little bias in their median.

Key words: Transient deformation; Rheology: crust and lithosphere; Continental tectonics: extensional; Satellite geodesy; Africa; Iceland

1 INTRODUCTION

The build-up of extensional strain between discrete rifting events is the least-constrained phase of the rifting cycle at divergent plate boundaries. Continental rifting drives the transition from stable lithosphere to oceanic spreading through rifting cycles that span a broad range of spatial and temporal scales (Corti 2009; Ebinger et al. 2013). These cycles alternate between discrete rifting events and the inter-rifting periods that separate them. Each event comprises two phases: elastic strain release through dike emplacement and normal faulting (Grandin et al. 2011; Maccaferri et al. 2016), followed by post-diking visco-elastic relaxation that decays on timescales of years to decades (Foulger et al. 1992; Nooner et al. 2009). Secular plate divergence, set by large-scale mantle dynamics (Saria et al. 2014; Stamps et al. 2021), persists throughout the cycle. It dominates strain accumulation during inter-rifting periods and is superposed on the rapid deformation of rifting events. Geodetic campaigns in Iceland (Heki et al. 1993; Hofton et al. 1995) and the Afar (Vigny et al. 2007; Nooner et al. 2009; Grandin et al. 2011) have documented the two event phases in detail.

Models of the inter-rifting phase conventionally borrow kinematic frameworks developed for the interseismic earthquake cycle. The standard approach adapts the backslip model of Savage & Prescott (1978): a normal fault bounding the rift is locked from the surface to a depth H and slips continuously at the plate rate below it. The elastic strain that accumulates on the locked segment then reproduces the observed near-field velocity gradient. Alternative formu-

lations represent the diking rift as a tensile dislocation in an elastic half-space that opens continuously at a depth-dependent rate (LaFemina et al. 2005; Konca et al. 2018), or prescribe a basal shear traction at the base of the plate to mimic far-field plate forcing (Ali et al. 2014). These models all ignore the rheological transition at the brittle–ductile boundary of magmatic rifts, where high heat flow and partial melt weaken the lower crust and uppermost mantle (Prodehl et al. 1997; Plasman et al. 2017). Geodetic and seismic constraints place this transition at 5–15 km depth in Iceland and the Afar (Stevens et al. 2021; Weinstein et al. 2017). There, the inferred asthenospheric viscosity ($\eta \approx 1$–5×10^{18} Pa s) implies Maxwell relaxation times of a few years (Hofton & Foulger 1996; Nooner et al. 2009), far shorter than the inter-rifting interval and incompatible with the purely elastic response assumed by backslip models. A continuously opening dislocation is also inconsistent with observations, because dike intrusions are finite, discrete events, and the geodetic record shows that each episode is followed by a localised transient decaying within a few years (Foulger et al. 1992; Reiss et al. 2022).

Analytical and numerical models of the post-diking transient itself meet a related obstacle. Analytical approaches, exemplified by the diffusion model of Foulger et al. (1992), reproduce the broad deformation pattern after individual intrusions, but they require an a priori viscous layer thickness, and their solutions return to a constant level far from the rift (Fig. 1). The observed profiles already capture the transition from post-diking relaxation to steady inter-rifting divergence. Twelve to seventeen years after the onset of rifting, the displacement field traces a broad, arctangent-like shape that extends well beyond the rift, and in the profile of Hofton & Foulger (1996) the two ends of the traverse settle onto distinct levels (Fig. 1b). In numerical visco-elastic simulations (Hofton et al. 1995; Hofton & Foulger 1996; Nooner et al. 2009), the lateral boundaries of the finite domain must either force the far-field velocity to zero or prescribe the divergence rate that the model should predict. A zero far-field velocity cannot be reconciled with the steady plate divergence observed in global reference frames (Fernandes et al. 2004; Altamimi et al. 2016). Neither class of model connects the transient post-diking signal to the secular strain rates of global plate motion models (Kreemer et al. 2014; Stamps et al. 2018).

The theoretical foundations laid for cyclic strike-slip fault deformation (Nur & Mavko 1974; Savage & Prescott 1978; Spence & Turcotte 1979; Cohen & Kramer 1984) motivate the view we adopt here. We treat inter-rifting deformation as the superposition of visco-elastic responses to an infinite history of discrete dike intrusions. The steadiness of inter-rifting velocities then results from the time-averaging of decaying transients, and requires no continuously creeping source.

We develop an analytical model, the elastic–viscous (EV) model, for an elastic plate overlying a viscous half-space (Hetland 2006) in which dike intrusions recur periodically at interval T and each open the rift by b, the dike opening. The governing integro-differential equation reduces, after a Fourier transform, to a fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993) parameterised by a single characteristic velocity $a = EH/[2\eta(1-\nu^2)]$. In the limit of many repeated intrusions, the system reaches a Recurrent Steady State (RSS) with a closed form in the Digamma and log-Gamma functions, whose two flanks diverge far from the rift at the long-term opening rate. We then extend the model to a Maxwell visco-elastic half-space, the elastic–visco-elastic (EVE) model, via the correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010), recovering two physically distinct relaxation regimes separated by the Maxwell timescale $\tau_R = \eta/\mu_m$. We apply both models to two geodetic datasets. At the Krafla rift, Iceland, horizontal displacement profiles were recorded 12–17 years after the onset of the 1975–1984 rifting sequence (Foulger et al. 1992; Hofton & Foulger 1996). At the Asal rift, Djibouti, a 25-year baseline time series is complemented by a rift-perpendicular velocity profile (Vigny et al. 2007). We use these records to constrain the viscosity of the half-space at both sites and to identify when a geodetic record can distinguish a viscous from a visco-elastic half-space.

2 ELASTIC–VISCOUS (EV) MODEL

2.1 Physical setup and governing equations

The model represents the lithosphere as a coupled mechanical system: a thin elastic plate of thickness H, which represents the brittle upper crust, overlying a Newtonian viscous half-space of viscosity η, which represents the ductile lower crust and uppermost mantle (e.g., Hetland & Hager 2005; Hetland 2006). The horizontal coordinate x is perpendicular to the rift axis; the vertical coordinate z is oriented downward, with the free surface at $z = 0$ and the base of the plate at $z = H$. We consider only the horizontal component of surface displacement; the vertical (uplift) field, although part of the geodetic record (Hofton & Foulger 1996; Nooner et al. 2009), lies outside the scope of this model. Under plane-strain conditions with negligible vertical stress ($\sigma_{zz} \approx 0$), the depth-averaged horizontal stress in the plate is related to the horizontal surface displacement $u(x,t)$ through Hooke's law:

$$\bar{\sigma}_{xx} = \frac{E}{1-\nu^2}\frac{\partial u}{\partial x}, \tag{1}$$

where E is Young's modulus and ν is Poisson's ratio. The factor $(1-\nu^2)^{-1}$ accounts for the lateral confinement of plane strain. Mechanical equilibrium of the plate requires the horizontal gradient of the depth-averaged stress to be balanced by the shear traction exerted at its base by the underlying half-space:

$$H\frac{\partial \bar{\sigma}_{xx}}{\partial x} + \sigma_{xz}^{\mathrm{BOT}} = 0. \tag{2}$$

The rheology of the half-space enters through the constitutive relation for the basal traction $\sigma_{xz}^{\mathrm{BOT}}$. For a Newtonian viscous half-space, the shear stress at x depends on the surface velocity at *all* other points through the non-local Stokes kernel (Savage & Prescott 1978):

$$\sigma_{xz}^{\mathrm{BOT}}(x,t) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\,\frac{\partial u}{\partial t}(x',t)\,dx', \tag{3}$$

interpreted as Hadamard finite part. This non-local coupling reflects the long-range character of viscous flow in the half-space: the traction induced at x by a velocity perturbation at x' decays only as $(x-x')^{-2}$.

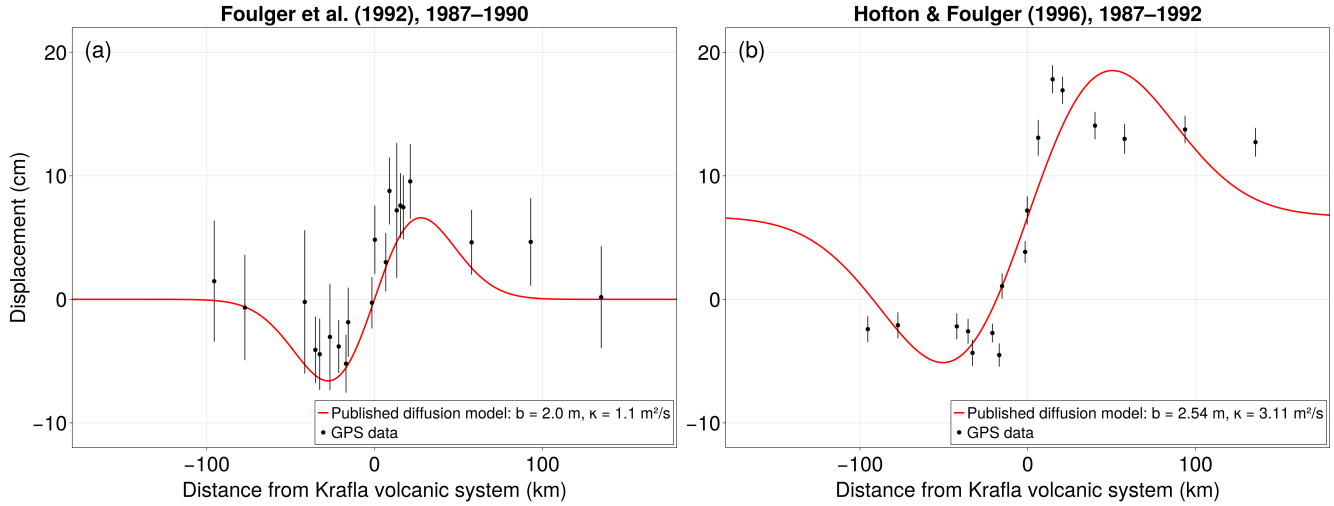


Figure 1. Post-rifting displacement profiles across the Krafla rift and the diffusion models published with them, reproduced without refitting. (a) Foulger et al. (1992), their Figure 2b: displacements of 20 stations between 1987 and 1990. (b) Hofton & Foulger (1996), their Figure 5: displacements of 17 stations between 1987 and 1992. Error bars are 1σ. Red curves: the best-fit diffusion model of each study, $u(x,t) = (b/2)\,\mathrm{erfc}[x/(2\sqrt{\kappa t})]$, written here in terms of the full dike opening b (both studies report the dike half-width $U_0 = b/2$). Following Foulger et al. (1992), each profile is modelled as the velocity of this solution at a time t after the intrusions, multiplied by the survey interval, with $t = 11$ yr in (a) and $t = 13$ yr in (b), as each study assumes. Published parameters: (a) $b = 2.0$ m ($U_0 = 1.0 \pm 0.1$ m) and $\kappa = 1.1 \pm 0.3$ m^2 s^{-1}; (b) $b = 2.54$ m ($U_0 = 1.27 \pm 0.14$ m) and $\kappa = 3.11 \pm 0.68$ m^2 s^{-1}. The zero reference of profile (b) is arbitrary (Hofton & Foulger 1996), so its curve is shifted by the constant 6.7 cm read from their Figure 16, not fitted.

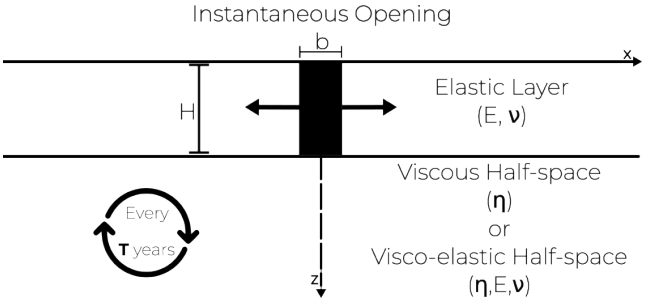


Figure 2. Model geometry. The elastic plate (thickness H, Young's modulus E, Poisson's ratio ν) overlies either a Newtonian viscous half-space (viscosity η) or, in its visco-elastic extension (Section 3.1), a Maxwell half-space of the same viscosity η that additionally shares the plate's elastic moduli E, ν – the EV and EVE models, respectively. Dike intrusions open the rift by b at intervals T; the horizontal surface displacement $u(x,t)$ is positive in the $+x$ direction. The parameter $a = EH/[2\eta(1-\nu^2)]$ sets the characteristic velocity that couples elastic and viscous deformation.

Substituting (1) into (2) and using (3) yields the governing integro-differential equation for u:

$$\frac{EH}{1-\nu^2}\frac{\partial^2 u}{\partial x^2} + \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}(x',t)\,dx' = 0. \qquad (4)$$

Dike intrusions enter as initial conditions: each event opens the rift instantaneously by b, offsetting the block on one side of the axis relative to the other (Fig. 2). Section 2.3 shows that this offset subsequently redistributes into a symmetric $\pm b/2$ split about the pre-intrusion position far from the rift, while the opening measured at the intrusion site itself remains exactly b at all times.

2.2 Fourier reduction and the characteristic velocity

Equation (4) is non-local in x and not tractable in physical space. A spatial Fourier transform resolves this. Defining the transform pair

$$\hat{u}(k,t) = \int_{-\infty}^{+\infty} u(x,t)e^{-ikx}\,dx,$$

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \hat{u}(k,t)e^{ikx}\,dk,$$

the transform of $\partial^2 u/\partial x^2$ is $-k^2\hat{u}$. The transform of the Stokes kernel convolution is:

$$\mathcal{F}\left\{\int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\frac{\partial u}{\partial t}\,dx'\right\} = -2\eta|k|\,\frac{\partial \hat{u}}{\partial t}. \qquad (5)$$

Applying the Fourier transform to (4) and using (5):

$$-\frac{EH}{1-\nu^2}k^2\hat{u} - 2\eta|k|\frac{\partial \hat{u}}{\partial t} = 0. \qquad (6)$$

Rearranging and introducing the characteristic velocity

$$a := \frac{EH}{2\eta(1-\nu^2)}, \qquad (7)$$

eq. (6) simplifies to a decoupled first-order ODE for each wavenumber k:

$$\frac{\partial \hat{u}}{\partial t} = -a\,|k|\,\hat{u}. \qquad (8)$$

The parameter a combines the elastic and viscous properties of the system: large a (stiff plate, low viscosity) implies rapid redistribution of post-diking strain over broad spatial scales; small a implies slow, locally confined relaxation. In physical space, the operator $|k|$ corresponds to the fractional

half-Laplacian $(-\Delta)^{1/2}$, so (8) is equivalently written as the fractional diffusion equation (Metzler & Klafter 2000; Samko & Ross 1993; Samko et al. 1993)

$$\frac{\partial u}{\partial t} = -a\,(-\Delta)^{1/2} u. \tag{9}$$

The solution of (8) for arbitrary initial data is:

$$\hat{u}(k,t) = \hat{u}(k,0)\, e^{-a|k|t}. \tag{10}$$

2.3 Step response to a single dike intrusion

The simplest building block of the model is the deformation produced by a single instantaneous dike intrusion at $t = 0$, which opens the rift by b. Because the governing eq. (4) constrains u only through its gradient, the reference level from which this offset is measured is arbitrary. We therefore write the initial condition, without loss of generality, as a Heaviside displacement, with the symmetric convention $\mathcal{H}(0) := 1/2$ so that the profile is well defined at the rift axis:

$$u(x, 0^+) = b\,\mathcal{H}(-x) = \begin{cases} 0\,, & x > 0 \\ b/2\,, & x = 0\,. \\ b\,, & x < 0 \end{cases} \tag{11}$$

Its Fourier transform, evaluated in the distributional sense in Appendix A, is

$$\hat{u}(k,0) = b\pi\delta(k) + \frac{ib}{k}. \tag{12}$$

Substituting (12) into (10) and noting that $\delta(k)$ is active only where $e^{-a|k|t} = 1$:

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k}\, e^{-a|k|t}. \tag{13}$$

The inverse transform of (13) reduces to a Dirichlet-type integral (Appendix A). We label quantities of this single-event solution with a STEP subscript, to distinguish them from the periodic RSS of Section 2.5. The step displacement is

$$u_{\mathrm{STEP}}(x,t) = \frac{b}{\pi} \arctan\!\left(\frac{at}{x}\right). \tag{14}$$

Differentiating with respect to t, the step velocity is a Lorentzian profile that broadens in time:

$$v_{\mathrm{STEP}}(x,t) = \frac{ba}{\pi}\, \frac{x}{x^2 + a^2 t^2}. \tag{15}$$

Both expressions depend on x and t only through a single combination, $x/(at)$: the self-similarity structure of the step response, developed in Section 2.4 below.

The surface velocity drives a flow throughout the underlying half-space. At depth z, the horizontal and vertical interior velocity components are the Poisson-kernel convolution of the surface velocity $v_s(x)$,

$$\begin{aligned} v_x(x,z) &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \sin^2\theta\, v_s(x - z\tan\theta)\, d\theta, \\ v_z(x,z) &= \frac{2}{\pi} \int_{-\pi/2}^{\pi/2} \sin\theta\cos\theta\, v_s(x - z\tan\theta)\, d\theta, \end{aligned} \tag{16}$$

with v_s set to the EV step velocity $v_{\mathrm{STEP}}(x,t)$ (15).

Figure 3 follows the relaxation of a single intrusion over three orders of magnitude in time, at the surface (left column) and in the underlying half-space (middle and right columns). At the surface, the response is a pulse of divergent velocity that travels away from the rift axis. Its peak sits exactly at $x = at$ (dashed lines), so a is the speed at which post-diking deformation propagates, and its amplitude decays as $b/(2\pi t)$. A point at distance $|x|$ has moved by $b/4$, half of its final share of the opening, when the peak passes it at $t = |x|/a$; the remaining $b/4$ accumulates only slowly, as u_{STEP} approaches $b/2$ as $1/t$. Ahead of the peak ($|x| \gg at$), eq. (15) reduces to the time-independent tail $ba/(\pi x)$, so a distant station records a nearly constant velocity from the moment of intrusion until the peak arrives. Behind the peak ($|x| \ll at$), the velocity falls as $bx/(\pi a t^2)$ and vanishes on the axis itself.

The interior field shows the flow that accompanies this surface pulse. Material rises beneath the axis and turns outward beneath the plate, beneath the diverging flanks (middle column). Because eq. (16) is linear in the surface velocity, the interior field inherits the self-similarity of eq. (15): t times the velocity depends only on $x/(at)$ and $z/(at)$, so the deforming region deepens at the same rate a as it widens. At distances much larger than at from the intrusion, the flow keeps its early, time-independent pattern, which is why the middle column barely changes across the six rows: even at $t = 3\tau_R$, $at \approx 4H$ remains small compared with the $30H$ window. Within at, the flow has already slowed down. The enlarged panels (right column) show this as a low-velocity region that opens beneath the axis and grows with t. The surface itself moves only horizontally ($v_z = 0$ at $z = 0$), consistent with the horizontal-only plate of Section 2.1.

2.4 Self-similarity of the step response

Both the displacement (14) and velocity (15) of Section 2.3 depend on x and t only through the self-similarity variable $\xi := x/(at)$, which measures distance in units of the velocity-peak position at. Dividing u_{STEP} by b and v_{STEP} by b/t collapses both onto universal curves, independent of E, H, η, or b:

$$\frac{u_{\mathrm{STEP}}}{b} = \frac{1}{\pi} \arctan\!\left(\frac{1}{\xi}\right), \tag{17}$$

$$\frac{t\, v_{\mathrm{STEP}}}{b} = \frac{1}{\pi}\, \frac{\xi}{1 + \xi^2}. \tag{18}$$

Figure 4a plots the displacement collapse (17) directly, alongside the Fickian Foulger et al. (1992) diffusion model: writing $u_{\mathrm{Foulger}}/b = \frac{1}{2}\,\mathrm{erfc}(\xi_F/2)$ in terms of its own self-similarity variable $\xi_F := x/\sqrt{\kappa t}$, both curves are shown for $\xi > 0$ only. This isolates the decay of the tails, a power law for EV and a Gaussian for Foulger, without reference to any particular parameter.

The same collapse applies to the velocity (18). Differentiating u_{Foulger}/b in t gives the Fickian model's velocity collapse:

$$\frac{t\, v_{\mathrm{Foulger}}}{b} = \frac{\xi_F}{4\sqrt{\pi}}\, e^{-\xi_F^2/4}. \tag{19}$$

Figure 4b plots (18) and (19) together: the EV velocity decays as the power law ξ^{-1}, a straight line on logarithmic axes, while the Fickian velocity decays as the far steeper

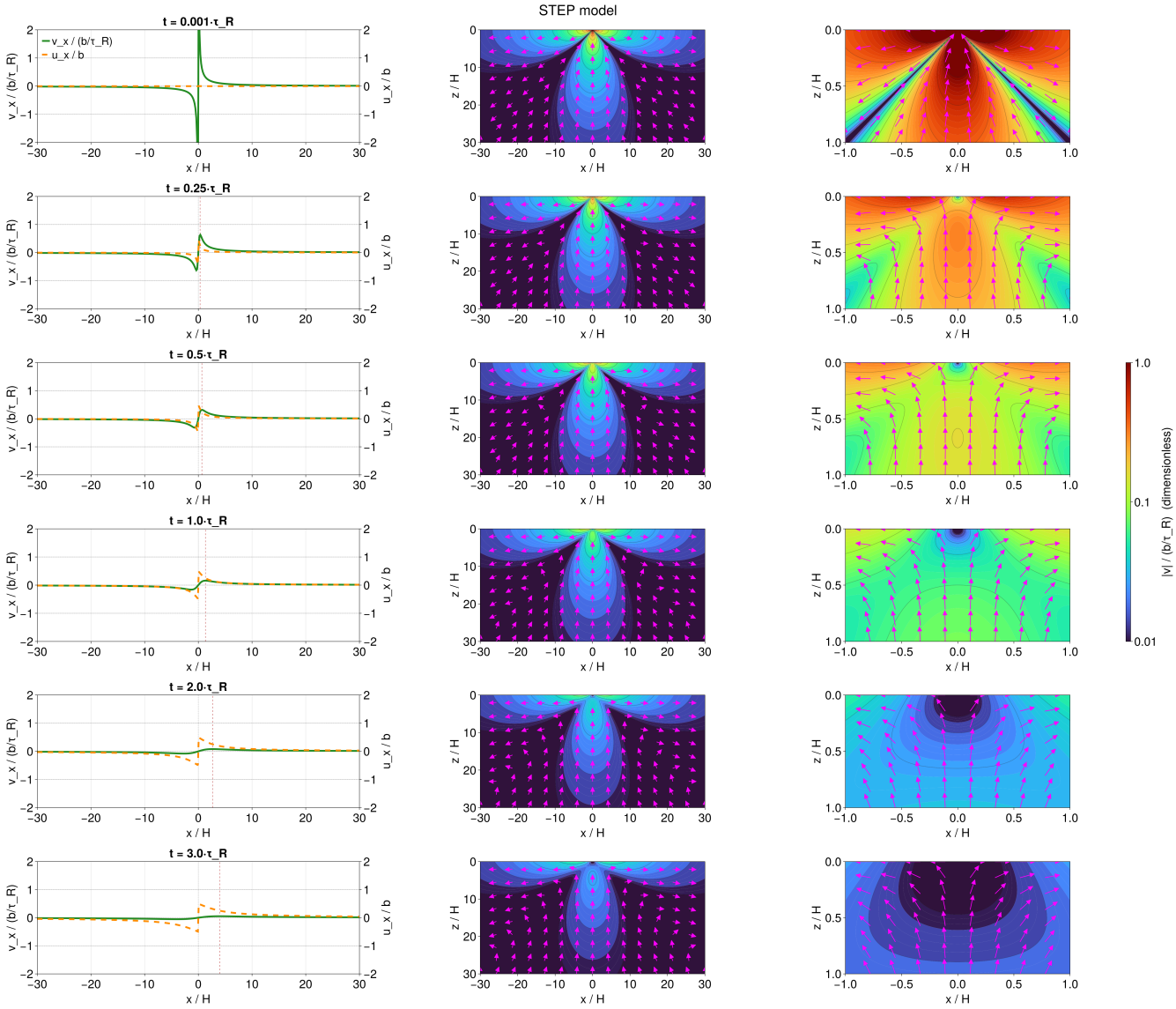


Figure 3. EV STEP response to a single dike intrusion of opening b at $t = 0$, at six times $t/\tau_R \in \{0.001, 0.25, 0.5, 1, 2, 3\}$, one per row. The Maxwell time $\tau_R = \eta/\mu_m$ (Section 3.1) serves here only as a time unit, the same one used for the EVE counterpart (Fig. 8). Left column: surface velocity v_{STEP} (eq. 15; solid, left axis, in units of b/τ_R) and displacement u_{STEP} (eq. 14; dashed, right axis, in units of b), counted from the instant after the intrusion, so that the two flanks move apart symmetrically and each tends to $\pm b/2$ at the axis. The dashed vertical line marks $x = at$, the position of the velocity peak. Middle column: speed $|v|$ in the half-space beneath the plate (filled contours, logarithmic scale, in units of b/τ_R), computed from the surface velocity with eq. (16) over $|x| \le 30H$ and $z \le 30H$. Arrows show the flow direction only and have a fixed length. Right column: the same field enlarged to $|x| \le H$, $z \le H$ beneath the rift axis, with the same colour scale. Speeds outside the colour bar range [0.01, 1] are clipped to its end colours. All lengths are scaled by the plate thickness H. Parameters: $H = 10$ km, $E = 40$ GPa, $\nu = 0.25$, $\eta = 5.2 \times 10^{18}$ Pa s, giving $a \approx 4.1 \times 10^{-5}$ m s^{-1} (≈ 1.3 km yr^{-1}) and $a\tau_R/H = 1/(1-\nu) \approx 1.33$.

$\xi_F\, e^{-\xi_F^2/4}$. Because ξ and ξ_F scale distance by different lengths, at and $\sqrt{\kappa t}$, the horizontal offset between the two curves carries no physical meaning, and only the shapes of the tails, a power law and a Gaussian, can be compared.

2.5 Cyclic accumulation and the Recurrent Steady State

2.5.1 Finite sum over past intrusions

Consider a rift that has hosted N dike intrusions at regular intervals T, the most recent a time $\tau \in [0, T)$ ago. Indexing the intrusions by $j = 0, 1, \ldots, N-1$, the number of cycles *before* the most recent one, the j-th intrusion has age $\tau + jT$. Superposing the step responses (14), each counted from the instant after its own intrusion, gives the accumulated

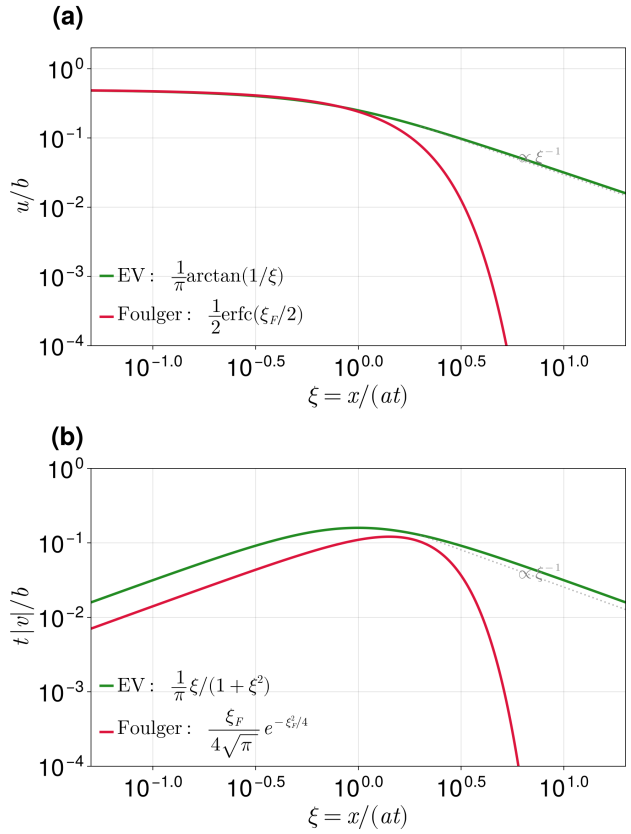


Figure 4. Self-similarity collapse of the step response in the EV model and in the Fickian Foulger et al. (1992) model, each plotted against its own self-similarity variable ($\xi = x/(at)$ for EV, $\xi_F = x/\sqrt{\kappa t}$ for Foulger) for $\xi > 0$. Both models start from the same opening b, so both displacements equal $b/2$ at the rift axis; neither curve is rescaled to a common length. (a) Displacement u/b: EV (17) and Foulger $\frac{1}{2}\mathrm{erfc}(\xi_F/2)$. (b) Velocity $t\,v/b$: EV (18) and Foulger (19). In both panels the dotted guide marks the ξ^{-1} power-law slope the EV curve approaches; the Fickian curves instead decay as a Gaussian.

displacement:

$$u(x,\tau) = \frac{b}{\pi}\sum_{j=0}^{N-1}\arctan\left(\frac{a(\tau+jT)}{x}\right). \tag{20}$$

Differentiating with respect to t, the corresponding velocity field is:

$$v(x,\tau) = \frac{ba}{\pi}\sum_{j=0}^{N-1}\frac{x}{x^2+a^2(\tau+jT)^2}. \tag{21}$$

2.5.2 *The RSS limit and Digamma representation*

After many cycles the system enters a periodic state, the rift analogue of the cyclic steady state described for the interseismic earthquake cycle (Thatcher & Rundle 1984; Hetland & Hager 2006). The RSS velocity is obtained by extending (21) to an infinite history by letting $N \to \infty$:

$$v_{\mathrm{RSS}}(x,\tau) = \frac{ba}{\pi}\sum_{j=0}^{\infty}\frac{x}{x^2+a^2(\tau+jT)^2}. \tag{22}$$

This series converges absolutely and sums to a closed form in the Digamma function $\psi(z)$ (Appendix B):

$$v_{\mathrm{RSS}}(x,\tau) = \frac{b}{\pi T}\,\mathrm{Im}\left\{\psi\left(\frac{\tau}{T}+i\frac{x}{aT}\right)\right\}. \tag{23}$$

2.5.3 *RSS displacement and far-field conservation*

Integrating (23) over τ from the most recent intrusion, and using the antiderivative $d[\ln\Gamma(z)]/dz = \psi(z)$, the RSS displacement accumulated since that intrusion is:

$$u_{\mathrm{RSS}}(x,\tau) = \frac{b}{\pi}\,\mathrm{Im}\left\{\ln\Gamma\left(\frac{\tau}{T}+i\frac{x}{aT}\right) - \ln\Gamma\left(i\frac{x}{aT}\right)\right\}. \tag{24}$$

Over one complete cycle, the cumulative displacement at any fixed $x > 0$ equals half the dike opening:

$$\int_0^T v_{\mathrm{RSS}}(x,\tau)\,d\tau = \frac{b}{2}, \qquad \forall\, x > 0. \tag{25}$$

This result follows from the Digamma representation: $\Gamma(1+iq) = iq\,\Gamma(iq)$ implies $\mathrm{Im}\{\ln\Gamma(1+iq) - \ln\Gamma(iq)\} = \pi/2$, giving the integral $b/2$. Each flank advances by half the dike opening per cycle, so the two flanks together separate by b, the opening of the dike intruded during that cycle. In the model, the relaxation of discrete intrusions thus suffices to sustain a steady far-field divergence at the rate measured by global plate motion models (Saria et al. 2014; Stamps et al. 2021), with no condition imposed at the lateral boundaries.

Figure 5 shows how this conservation plays out over successive cycles. After n complete cycles and a further phase τ, the displacement accumulated since a reference intrusion is $n(b/2)\,\mathrm{sgn}(x) + u_{\mathrm{RSS}}(x,\tau)$. At the end of each cycle, $u_{\mathrm{RSS}}(x,T^-) = b/2$ at every $x > 0$ (eq. 25), so each flank has translated as a rigid block by $b/2$ and the profile is uniform (thick lines in Fig. 5a). Within the cycle, the region next to the axis takes up its share of the opening almost immediately after the intrusion, while distant points catch up later in the cycle. The resulting histories (Fig. 5b) range from a staircase at the axis, which jumps by $b/2$ at each intrusion, to a nearly uniform rate $b/(2T)$ far from it. At $x = 0.1\,aT$, nearly 80 per cent of the $b/2$ is gained in the first quarter of the cycle; at $x = aT$, the history departs from the uniform rate by at most $0.04\,b$, and at $x = 3\,aT$ by at most $0.013\,b$. A station at $|x| \gtrsim aT$ therefore records an almost steady velocity, the inter-rifting plate divergence.

The interior flow of the RSS follows from the same Poisson-kernel extension (16), with v_s now set to the EV RSS velocity $v_{\mathrm{RSS}}(x,\tau)$ (23). Figure 6 shows it in the same layout as the STEP field of Section 2.3 (Fig. 3), for $T = 5\tau_R$ and at six phases spread over one cycle: the surface profile (left column), the speed in the half-space (middle column), and an enlargement beneath the axis (right column). Distances are scaled by aT and velocities by the rift opening rate $V = b/T$, so colours are not comparable with those of the STEP figure, whose unit is b/τ_R.

Unlike the STEP velocity, the RSS velocity does not decay from one cycle to the next. Far from the axis ($|x| \gg aT$), each flank moves at $V/2$ throughout the cycle, so the two flanks separate at the full rift opening rate V, as required by eq. (25). Near the axis, each cycle repeats the behaviour of the single event. Immediately after an intrusion, the near-axis velocity exceeds V by more than an order of magnitude, and a velocity peak migrates away from the axis close to $x =$

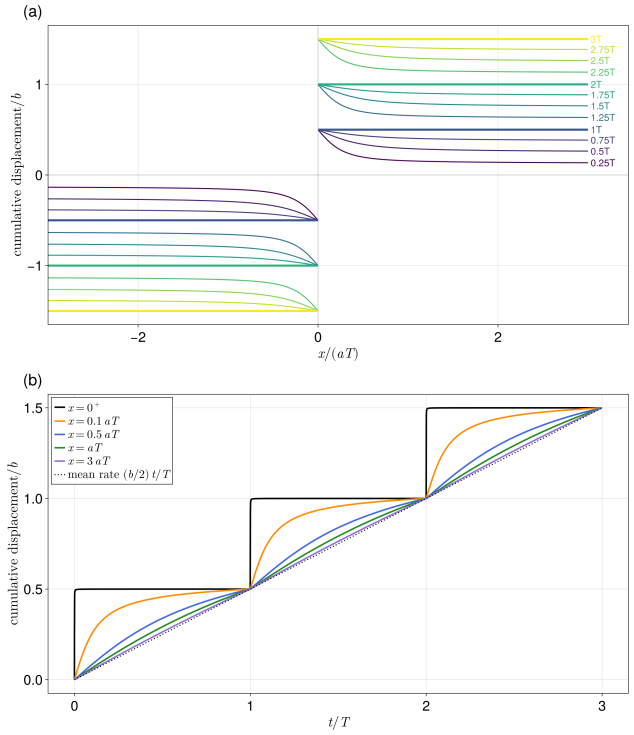


Figure 5. Displacement accumulated in the EV RSS over three cycles, starting from an intrusion at $t = 0$ that follows an infinite periodic history. Displacement is normalised by b, distance by aT and time by T. (a) Profiles every $T/4$; thick lines mark the end of each cycle, when both flanks have moved uniformly by $b/2$ per cycle. (b) Displacement histories at fixed distances from the axis ($x > 0$), from the staircase at $x = 0^+$ to the near-uniform rate at $x = 3\,aT$; the dotted line marks the mean rate $(b/2)\,t/T$.

$a\tau$ (dashed lines), as in Section 2.3. By the end of the cycle, the near-axis velocity has fallen well below $V/2$, and the profile has smoothed into a monotonic ramp from $-V/2$ to $+V/2$ over a distance of order aT. The interior field follows the same pattern. Upwelling beneath the axis and outward flow beneath the plate persist at every phase, and over most of the half-space the speed stays between roughly $0.1\,V$ and $V/2$ throughout the cycle (middle column). Only the region beneath the axis changes. As in the STEP response, a low-velocity region opens there after each intrusion and grows until the next one (right column).

Equation (23) also fixes the scales of the RSS. Normalised by the rift opening rate $V = b/T$, the velocity depends only on the phase τ/T and on $x/(aT)$, which together form the Digamma argument $\tau/T + ix/(aT)$: aT, which we call the deformation half-width, sets the spatial scale of the deformation zone and b/T its amplitude. Late in the cycle, aT is the distance over which the velocity rises from zero at the axis to the far-field value $\pm V/2$; early in the cycle, the near-axis profile is dominated by the most recent intrusion, whose velocity peak sits at $x \approx a\tau < aT$ (Fig. 6). Because $aT \propto EHT/\eta$, a geodetic profile that resolves this scale constrains EH/η wherever T can be estimated independently.

The RSS assumes an infinite history of intrusions. Truncating the sum (21) after N intrusions removes the contribution of the older ones; approximating the missing terms by an integral, the truncated velocity falls short of the RSS by $\approx (V/\pi)\arctan[|x|/(NaT)]$ (Fig. 7). After ten cycles, the deficit is 6 per cent of the far-field velocity $V/2$ at $|x| = aT$ but 50 per cent at $|x| = 10\,aT$, and the truncated sum stays within 5 per cent of $V/2$ only where $N \gtrsim 13\,|x|/(aT)$. Physically, the far-field plateau $\pm V/2$ is carried by old intrusions, whose single-event velocity peaks have migrated to $x \approx a(\tau + jT)$; a station at distance $|x|$ records the RSS only if the rift has hosted intrusions for much longer than $|x|/a$. Where this condition fails, as on a newly initiated rift segment, the explicit sum (20) should replace the RSS.

3 ELASTIC–VISCO-ELASTIC (EVE) MODEL

3.1 Maxwell half-space and the visco-elastic Stokes kernel

The Maxwell relaxation time $\tau_R := \eta/\mu_m$, where μ_m is the shear modulus of the half-space, separates two regimes: on timescales $t \ll \tau_R$ the half-space responds elastically, and on timescales $t \gg \tau_R$ it flows viscously (Bürgmann & Dresen 2008). We model this by replacing the Newtonian half-space with a Maxwell body, a Hookean spring and a Newtonian dashpot in series. By the elastic–visco-elastic correspondence principle (Hetland & Hager 2005; Barbot & Fialko 2010), a Maxwell body responds in the Laplace domain like a dashpot with the frequency-dependent effective viscosity (Appendix C)

$$\tilde{\eta}(s) = \frac{1}{s/\mu_m + 1/\eta} = \frac{\eta}{1 + \tau_R s}. \tag{26}$$

At $s \to 0$ (long times) $\tilde{\eta}(s) \to \eta$, recovering the bare Newtonian dashpot; at $s \to \infty$ (short times) $\tilde{\eta}(s)\,s \to \mu_m$, recovering the purely elastic response of the spring. Replacing η by $\tilde{\eta}(s)$ in the Stokes kernel (3) and returning to the time domain gives the visco-elastic Stokes kernel (Appendix C):

$$\sigma_{xz}^{\mathrm{BOT}} + \tau_R \frac{\partial \sigma_{xz}^{\mathrm{BOT}}}{\partial t} = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi (x - x')^2} \frac{\partial u(x', t)}{\partial t}\, dx'. \tag{27}$$

In the limit $\tau_R \to 0$, (27) reduces to the EV Stokes kernel (3).

3.1.1 Governing equation in Fourier space

The equilibrium eq. (2) expresses the basal traction in terms of the plate displacement:

$$\sigma_{xz}^{\mathrm{BOT}} = -\frac{EH}{1 - \nu^2} \frac{\partial^2 u}{\partial x^2}. \tag{28}$$

Substituting (28) and its time derivative into (27), then applying the spatial Fourier transform using (5):

$$\frac{EH}{1 - \nu^2} k^2 \hat{u} + \tau_R \frac{EH}{1 - \nu^2} k^2 \frac{\partial \hat{u}}{\partial t} = -2\eta |k| \frac{\partial \hat{u}}{\partial t}. \tag{29}$$

Dividing by 2η, using (7), and collecting $\partial\hat{u}/\partial t$ terms gives the visco-elastic analogue of (8):

$$\frac{\partial \hat{u}}{\partial t} = -\left(\frac{a|k|}{1 + \tau_R a |k|} \right) \hat{u}. \tag{30}$$

The denominator $1 + \tau_R a|k|$ caps the relaxation rate at $1/\tau_R$ for $|k| \gg 1/(a\tau_R)$, so short-wavelength (near-field) features decay more slowly than in the EV case.

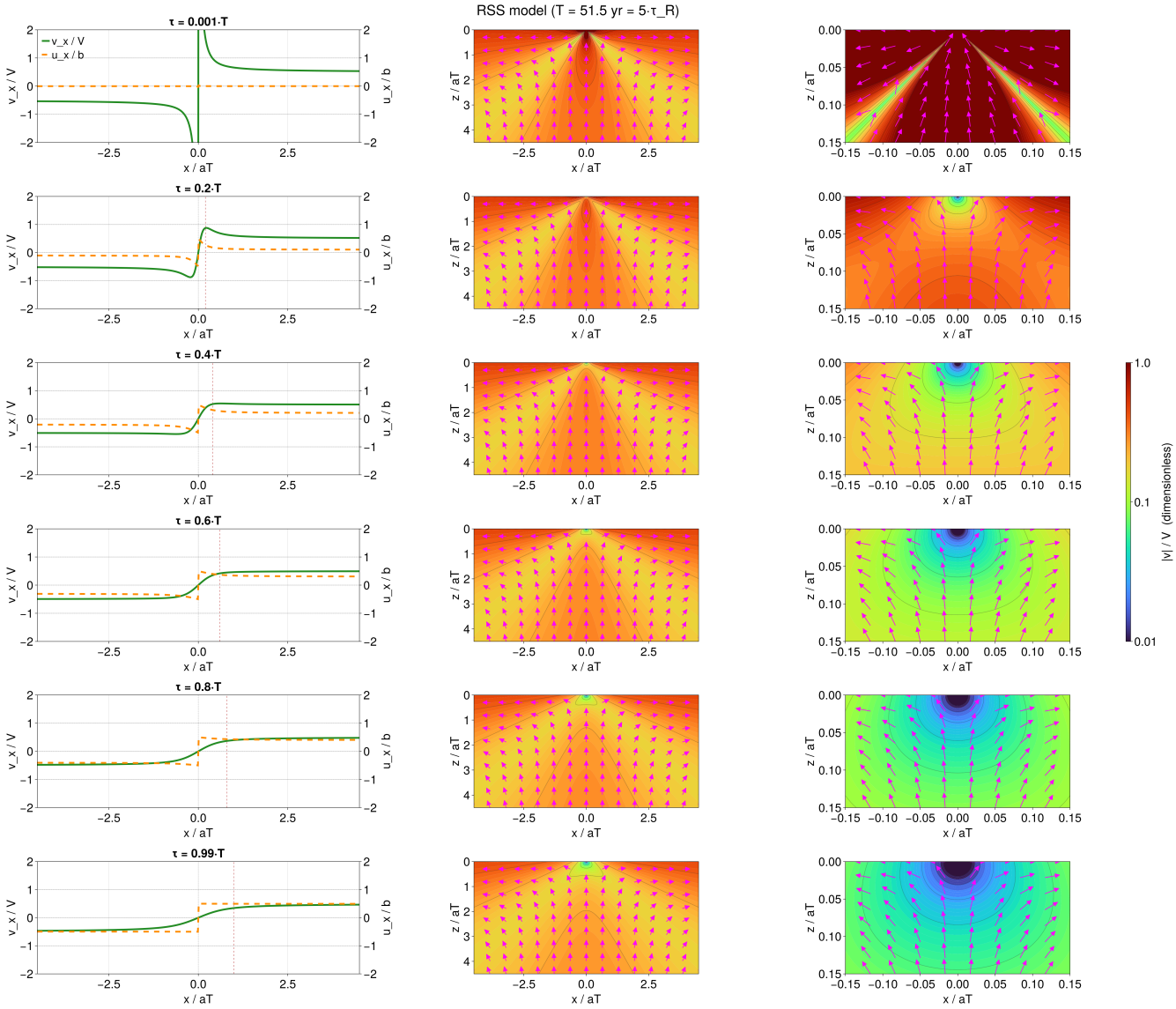


Figure 6. EV RSS for $T = 5\tau_R$, in the same layout as the STEP response (Fig. 3), at six phases $\tau/T \in \{0.001, 0.2, 0.4, 0.6, 0.8, 0.99\}$, one per row. Left column: surface velocity $v_{\rm RSS}$ (eq. 23; solid, left axis, in units of $V = b/T$) and displacement accumulated since the most recent intrusion $u_{\rm RSS}$ (eq. 24; dashed, right axis, in units of b). The dashed vertical line marks $x = a\tau$, the velocity-peak position of the most recent intrusion alone. Middle column: speed $|v|$ in the half-space (filled contours, logarithmic scale, in units of V), computed with eq. (16) over $|x| \le 4.5\,aT$ and $z \le 4.5\,aT$. Arrows show the flow direction only and have a fixed length. Right column: the same field enlarged to $|x| \le 0.15\,aT$, $z \le 0.15\,aT$ beneath the rift axis, with the same colour scale. Speeds outside the colour bar range [0.01, 1] are clipped to its end colours. Parameters as in Fig. 3, giving $aT \approx 67$ km.

3.2 Step response and asymptotic behaviour

Equation (30) is separable in k, yielding a decaying exponential for each wavenumber:

$$\hat{u}(k,t) = \hat{u}(k,0)\exp\left(-\frac{a|k|}{1+\tau_R a|k|}\,t\right). \tag{31}$$

Applying the Heaviside initial condition (12):

$$\hat{u}(k,t) = b\pi\delta(k) + \frac{ib}{k}\exp\left(-\frac{a|k|}{1+\tau_R a|k|}\,t\right). \tag{32}$$

Taking the inverse Fourier transform and following the same parity argument as in Section 2.3, the EVE step displacement requires numerical integration:

$$u^{\rm EVE}_{\rm STEP}(x,t) = \frac{b}{2} - \frac{b}{\pi}\int_0^{+\infty}\frac{\sin(kx)}{k}\exp\left(-\frac{akt}{1+\tau_R ak}\right)dk, \tag{33}$$

since the rational exponent prevents the closed-form Dirichlet evaluation of Appendix A. The surface velocity follows by differentiating (33) with respect to t:

$$v^{\rm EVE}_{\rm STEP}(x,t) = \frac{ba}{\pi}\int_0^{+\infty}\frac{\sin(kx)}{1+\tau_R ak}\exp\left(-\frac{akt}{1+\tau_R ak}\right)dk. \tag{34}$$

We evaluate both integrals numerically (Appendix D).

The exponent in (33) reveals two physically distinct regimes. In the long-wavelength limit ($k \to 0$ or $t \to \infty$),

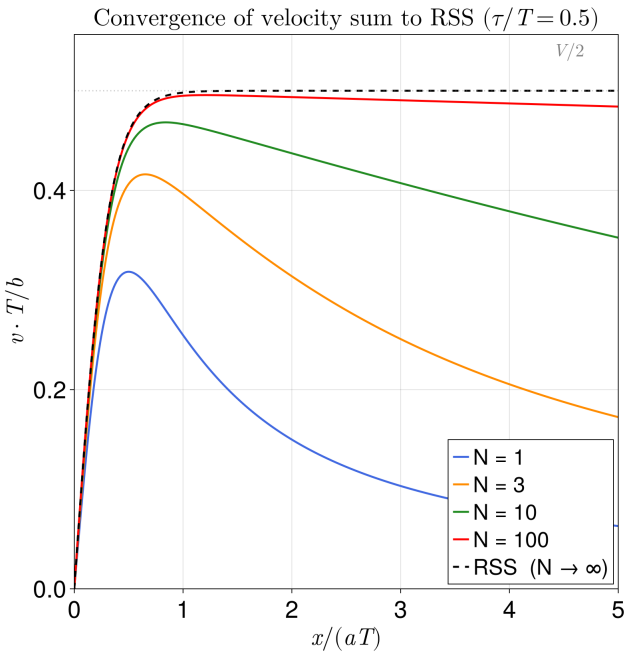


Figure 7. Convergence of the finite-sum velocity (21) to the RSS (23) at phase $\tau/T = 0.5$, for $N = 1$, 3, 10 and 100 past intrusions (solid curves) and the RSS (dashed). Velocity is normalised by b/T and distance by aT. The deficit of the truncated sum grows with distance, approximately as $(b/\pi T)\arctan[x/(NaT)]$: with $N = 10$, the sum matches the RSS near the axis but remains below the far-field plateau $b/(2T)$ (dotted) for $x \gtrsim aT$.

$1 + \tau_R ak \approx 1$ and the exponential reduces to $\exp(-akt)$, recovering the EV Dirichlet integral of Section 2.3. The elastic memory of the half-space fades at large distances and late times, and the model converges to the Newtonian viscous limit. In the short-wavelength limit ($k \to \infty$ or $t \to 0$), $1 + \tau_R ak \approx \tau_R ak$ and the exponent saturates:

$$\lim_{k\to\infty}\left(-\frac{akt}{1+\tau_R ak}\right) = -\frac{t}{\tau_R}. \tag{35}$$

The saturation applies to wavenumbers $k \gtrsim 1/\ell_M$, where $\ell_M := a\tau_R$ is the Maxwell length. At early times $t \ll \tau_R$, the exponential is close to unity at all wavenumbers, so the displacement remains close to the initial step. For $t \sim \tau_R$, near-field deformation decays exponentially at the fixed rate $1/\tau_R$, representing pure Maxwell relaxation of the half-space independent of the plate. Together, (34) and (35) show that the EVE model introduces a short-time, near-field stiffening relative to the EV model, and that the EVE step response converges to the EV one at $t \gg \tau_R$.

The saturated exponent also gives the near-axis velocity in closed form. For $k \gg 1/(a\tau_R)$, the integrand of (34) reduces to $\sin(kx)\, e^{-t/\tau_R}/(\tau_R k)$, and these wavenumbers control the velocity as $x \to 0$. The Dirichlet integral of Section 2.3 then gives

$$v^{\mathrm{EVE}}_{\mathrm{STEP}}(x \to 0^+,\, t) = \frac{b}{2\tau_R}\, e^{-t/\tau_R}, \tag{36}$$

a finite plateau that decays at the Maxwell rate, whereas the EV velocity vanishes at the axis. The near-field signal of the EVE model is an elastic rebound of the half-space.

Figure 8 shows the EVE STEP response in the same layout as its EV counterpart (Fig. 3): same velocity unit b/τ_R, colour scale, windows, vertical guide $x = at$ and times, except that its first row shows the limit $t \to 0^+$ rather than $t = 0.001\,\tau_R$. Differences between the two figures reflect only the rheology. Unlike the EV case, the surface velocity is finite immediately after the intrusion and reaches $b/(2\tau_R)$ at the axis, the near-field plateau of eq. (36). Peak surface velocity is largest at $t \to 0^+$ and falls with increasing t, and the underlying flow lobe shrinks and thins in step, the same qualitative decay as the EV STEP field, but on the timescale τ_R.

3.3 EVE Recurrent Steady State

3.3.1 Infinite sum and Fourier periodicity condition

The EVE RSS is the superposition of step-response velocities from all preceding intrusions, by the same argument as in Section 2.5:

$$v^{\mathrm{EVE}}_{\mathrm{RSS}}(x,\tau) = \sum_{j=0}^{\infty} v^{\mathrm{EVE}}_{\mathrm{STEP}}(x,\, \tau + jT). \tag{37}$$

Writing $v^{\mathrm{EVE}}_{\mathrm{STEP}}$ in Fourier space (Section 3.1, eq. (34)) as $\frac{ba}{\pi}\int_0^\infty \frac{\sin(kx)}{1+\tau_R ak}\, e^{-\alpha(k)(\tau+jT)}\, dk$, with the effective Maxwell decay rate

$$\alpha(k) := \frac{ak}{1+\tau_R ak}, \tag{38}$$

each past intrusion contributes a factor $e^{-\alpha(k)jT}$. The series converges for all $k > 0$ and the geometric sum $\sum_{j=0}^{\infty} e^{-\alpha(k)jT} = (1 - e^{-\alpha(k)T})^{-1}$ can be performed inside the Fourier integral. The same result follows from requiring each Fourier mode to repeat from one cycle to the next, apart from the offset added by each intrusion, $\hat{u}(k, 0^+) = \hat{u}(k, T^-) + ib/k$. Together with the decay (30) within the cycle, this condition has the unique solution

$$\hat{u}(k,\tau) = \frac{ib/k}{1 - e^{-\alpha(k)T}}\, e^{-\alpha(k)\tau}, \qquad \tau \in [0, T). \tag{39}$$

3.3.2 Physical-space representation, conservation, and limits

Inverting (39) gives the EVE RSS velocity and displacement:

$$v^{\mathrm{EVE}}_{\mathrm{RSS}}(x,\tau) = \frac{ba}{\pi}\int_0^\infty \frac{\sin(kx)}{1+\tau_R ak}\,\frac{e^{-\alpha(k)\tau}}{1 - e^{-\alpha(k)T}}\, dk, \tag{40}$$

$$u^{\mathrm{EVE}}_{\mathrm{RSS}}(x,\tau) = \frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\,\frac{1 - e^{-\alpha(k)\tau}}{1 - e^{-\alpha(k)T}}\, dk. \tag{41}$$

Integrating the velocity over one full period, using $\int_0^T e^{-\alpha\tau}/(1 - e^{-\alpha T})\, d\tau = 1/\alpha$ and the algebraic identity $a/[(1 + \tau_R ak)\,\alpha(k)] = 1/k$, gives

$$\int_0^T v^{\mathrm{EVE}}_{\mathrm{RSS}}(x,\tau)\, d\tau = \frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\, dk = \frac{b}{2}, \tag{42}$$

where the last step is the Dirichlet integral. Half the dike opening, $b/2$, is recovered every cycle, independent of τ_R, matching the EV RSS result (25) and the late-time limit $b/2$ of the single-event EVE displacement (33). In the limit $\tau_R \to 0$, $\alpha(k) \to ak$ and $(1 + \tau_R ak)^{-1} \to 1$, and (40) reduces to the EV RSS Fourier integral whose closed form is

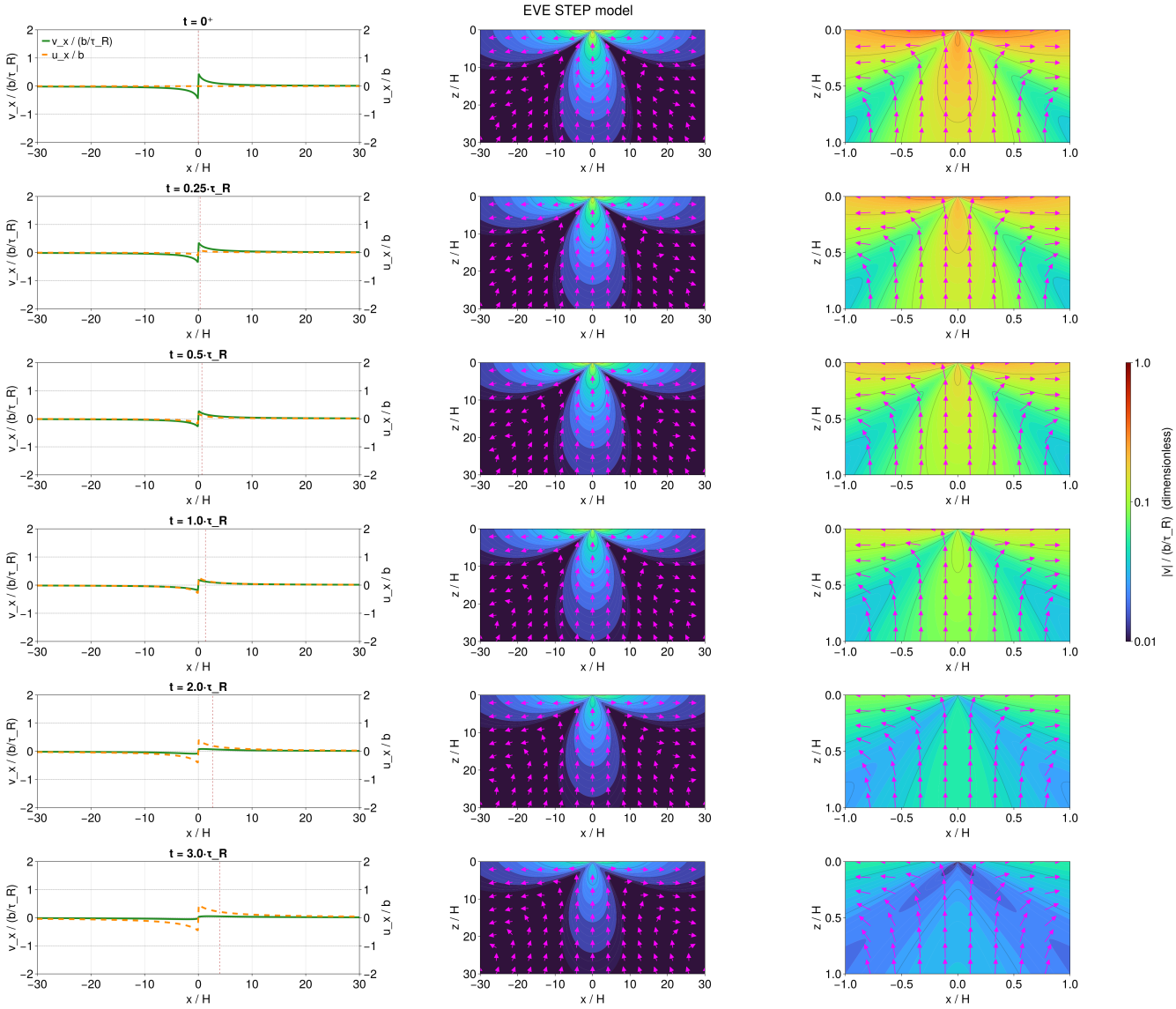


Figure 8. EVE STEP response to a single dike intrusion of opening b. Layout, units, colour scale and parameters as in Fig. 3, with $\tau_R = \eta/\mu_m$ from the shared-modulus closure (eq. 47). Left column: $v_{\mathrm{STEP}}^{\mathrm{EVE}}$ (eq. 34) and $u_{\mathrm{STEP}}^{\mathrm{EVE}}$ (eq. 33). Rows show $t/\tau_R \in \{0^+, 0.25, 0.5, 1, 2, 3\}$; the first row is the limit $t \to 0^+$, evaluated in closed form, where the surface velocity reaches $b/(2\tau_R)$ at the axis (eq. 36). The dashed vertical line marks the EV velocity-peak position $x = at$.

the Digamma series of Section 2.5. For $\tau_R > 0$, the saturating denominator $1 + \tau_R ak$ prevents this reduction, and we evaluate (40) numerically (Appendix D).

The interior flow of the EVE RSS again follows from eq. (16), with v_s set to $v_{\mathrm{RSS}}^{\mathrm{EVE}}$ (40). Figure 9 shows it in the same layout, normalisation and phases as the EV RSS (Fig. 6), for the same $T = 5\tau_R$, except that its first row shows the limit $\tau \to 0^+$. Far from the axis ($|x| \gg aT$), each flank moves at $V/2$ throughout the cycle, as in the EV RSS. Near the axis, the cycle starts from a finite velocity jump at the source. Each past intrusion contributes the plateau (36) at its own age $\tau + jT$, and summing these contributions gives

$$v_{\mathrm{RSS}}^{\mathrm{EVE}}(x \to 0^+, \tau) = \frac{b}{2\tau_R} \frac{e^{-\tau/\tau_R}}{1 - e^{-T/\tau_R}}, \qquad (43)$$

so the velocity jumps by twice this value across the axis for any $\tau_R > 0$, whereas the EV RSS vanishes continuously there. Immediately after an intrusion, the near-axis velocity is about $2.5\,V$ (eq. 43), the elastic rebound of the most recent intrusion together with the remnants of earlier ones. This jump decays at the rate $1/\tau_R$, to about $0.9\,V$ at $\tau = 0.2\,T = \tau_R$ and $0.3\,V$ at $\tau = 0.4\,T$. By the end of the cycle it has all but vanished, and the profile has smoothed into a monotonic ramp from $-V/2$ to $+V/2$ over a distance of order aT, as in the EV RSS. The interior field follows the same pattern as in the EV case. Upwelling beneath the axis and outward flow beneath the plate persist at every phase, and the speed over most of the half-space stays between roughly $0.1\,V$ and $V/2$ (middle column). A low-velocity region opens beneath the axis after each intrusion and grows until the next one (right column).

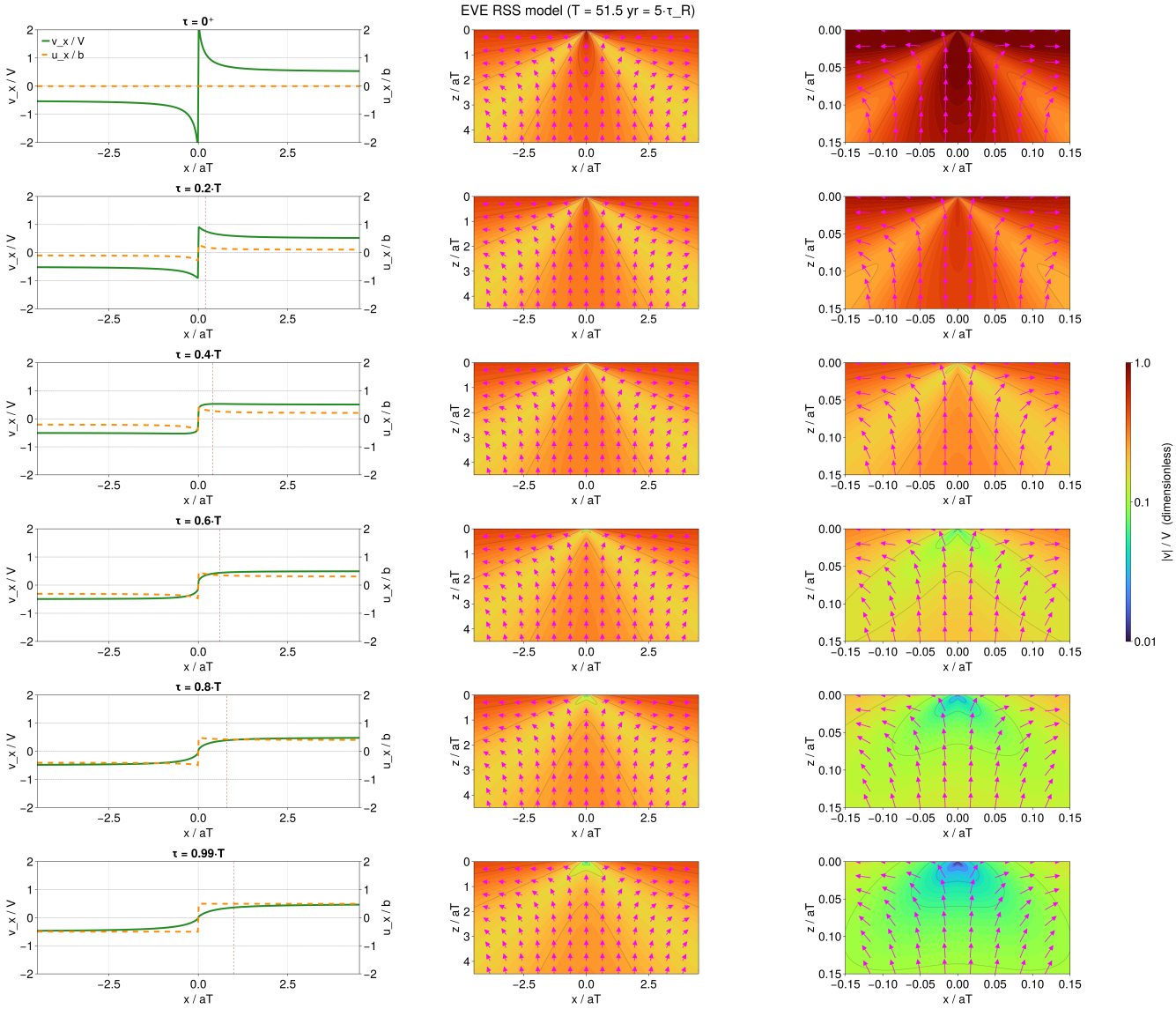


Figure 9. EVE RSS for $T = 5\tau_R$. Layout, units, colour scale, phases and parameters as in Fig. 6, with $\tau_R = \eta/\mu_m$ from the shared-modulus closure (eq. 47). Left column: $v_{\rm RSS}^{\rm EVE}$ (eq. 40) and $u_{\rm RSS}^{\rm EVE}$ (eq. 41). The first row is the limit $\tau \to 0^+$, computed from its exact form: the displacement since the most recent intrusion is zero, and the axial velocity $(b/2\tau_R)/(1 - e^{-T/\tau_R}) \approx 2.5\,V$ (eq. 43) lies beyond the plotted range and is clipped at $2\,V$. The dashed vertical line marks the EV velocity-peak position $x = a\tau$ of the most recent intrusion.

3.4 Shared-modulus closure for τ_R

Section 4 fits the EVE RSS to real data alongside its EV counterpart. A free τ_R alongside (a, b) would let the extra parameter absorb noise rather than test whether the EVE model improves the fit. A model with one more free parameter always reaches at least as low a χ^2, whether or not that parameter is physically warranted. We instead derive τ_R from a, so that the EVE RSS carries the same number of free parameters as the EV RSS it is compared against.

Doing so requires the shear modulus μ_m of the half-space, left general in eq. (26) so far. By definition, $\tau_R := \eta/\mu_m$ is a material property of the half-space alone. The EV model already contains an analogous timescale, the ratio of the half-space viscosity to the shear modulus of the plate,

$$\tau_{\rm EV} := \frac{\eta}{\mu_{\rm plate}}, \qquad \mu_{\rm plate} := \frac{E}{2(1+\nu)}, \tag{44}$$

which is the time the EV velocity peak takes to travel $a\tau_{\rm EV} = H/(1-\nu) \approx 1.3\,H$. The two timescales differ only in which shear modulus, μ_m or $\mu_{\rm plate}$, is paired with η.

We adopt the simplest closure available without additional data: the half-space shares the shear modulus of the elastic plate it underlies,

$$\mu_m = \mu_{\rm plate}. \tag{45}$$

This closure makes the two relaxation times identical:

$$\tau_R = \frac{\eta}{\mu_m} = \frac{\eta}{\mu_{\rm plate}} = \tau_{\rm EV}. \tag{46}$$

Combined with $\eta = EH/[2a(1-\nu^2)]$, the inversion of the

characteristic velocity (7), this gives

$$\tau_R = \tau_{\mathrm{EV}} = \frac{H}{a(1-\nu)}, \tag{47}$$

a relaxation time set entirely by the fitted characteristic velocity a and the canonical parameters H and ν (Section 5.4), with no independent estimate of μ_m or η. The shared-modulus closure is a simplifying assumption, because the shear modulus of the half-space can differ from that of the plate. We adopt it in the absence of independent constraints, and it keeps the EV-versus-EVE comparison of Section 4 an equal-parameter test of functional form.

4 APPLICATIONS TO MAGMATIC RIFT SYSTEMS

We test the EV and EVE Recurrent Steady States (Sections 2.5 and 3.3) against post-diking geodetic records from two magmatic rifts. The first is the 1975–1984 Krafla rifting episode in NE Iceland, sampled by two rift-perpendicular displacement profiles (Foulger et al. 1992; Hofton & Foulger 1996). The second is the 1978 Asal–Ghoubbet episode in the Afar Triangle, sampled by a 25-year baseline time series and a rift-perpendicular velocity profile (Vigny et al. 2007). The two sites differ in rift opening rate, in the length of their records and in the type of data, so together they test whether a single functional form describes inter-rifting deformation in contrasting settings.

Both records cover only part of one cycle. They span 12–17 yr after the onset of the Krafla episode and 1–25 yr after the Asal event, and in neither case has a later intrusion been documented. We set the recurrence interval through the far-field conservation of the RSS (eq. 25). Each flank advances by $b/2$ per cycle, so the two flanks separate by the full dike opening b every T. The rift opening rate is then $V = b/T$, and each flank moves at $\pm V/2$ far from the axis. We take V from an independent, long-term estimate of the rift opening rate, from global plate motion models or regional geodetic studies, and derive the recurrence interval from the fitted opening as $T = b/V$. This leaves two free parameters, the characteristic velocity a and the dike opening b, plus a reference-frame offset where the data require one. The EVE RSS carries τ_R derived from a through the shared-modulus closure (eq. 47). Each fit uses the full RSS, which includes the relaxation of all earlier intrusions and the far-field plate motion they sustain.

At each site we minimise the misfit

$$\chi^2 = \sum_{i=1}^{n} \left(\frac{u_i^{\mathrm{obs}} - u_i^{\mathrm{pred}}}{\sigma_i} \right)^2, \tag{48}$$

where u_i^{obs} and u_i^{pred} are the observed and predicted displacement, baseline change or velocity at data point i, σ_i is its 1σ uncertainty, and n is the number of observations. We report the reduced statistic $\chi^2_r = \chi^2/(n-p)$, where p is the number of free parameters, so that $\chi^2_r \approx 1$ indicates a fit within measurement noise. We compare the EV and EVE fits with the Akaike information criterion (AIC) and report $\Delta\mathrm{AIC} = \mathrm{AIC} - \mathrm{AIC}_{\mathrm{EV}}$, so that positive values favour the EV RSS. Because both models carry the same number of free parameters, the AIC difference reduces to the difference in χ^2, and any preference reflects only the shape of the predicted deformation.

4.1 Krafla rift, Iceland

The Krafla rifting episode began in December 1975 and lasted about a decade, with some 20 dike intrusions along the Krafla fissure swarm (Hofton & Foulger 1996). Its aftermath was recorded by GPS campaigns in 1987, 1990 and 1992. We digitised two rift-perpendicular displacement profiles from these campaigns. The first comprises 20 stations of Foulger et al. (1992) (their Figure 2b), differenced between 1987 and 1990. The second comprises 17 stations of Hofton & Foulger (1996) (their Figure 5, the same profile they compare with the diffusion model in their Figure 16), differenced between 1987 and 1992. Each point carries its published 1σ error bar. The two profiles also differ in their spatial reference. Foulger et al. (1992) tie their displacements to near-rift stations that showed no significant deformation, whereas Hofton & Foulger (1996) state that the zero level of their profile is arbitrary. We therefore add a constant offset c to the model for the Hofton & Foulger (1996) profile alone and solve for it during the fit.

We measure time for both profiles from the onset of the episode, $t_0 = 1975$. Our RSS represents each episode as a single intrusion of opening b and predicts each profile from the elapsed times (τ_i, τ_f) since that intrusion, so the two datasets need a common origin. Neither published study measures time from the onset. Foulger et al. (1992) treat their 1987–1990 displacements as an instantaneous velocity 11 yr after the dike intrusions, and Hofton & Foulger (1996) treat their 1987–1992 displacements as one 13 yr after, so both place the intrusion in 1979. Hofton & Foulger (1996) likewise inject all dikes of their visco-elastic model simultaneously in 1979, and report that spreading the injections over the real nine-year sequence changes the 1987–1992 field negligibly. We prefer the onset because the half-space begins to relax as soon as the first dike opens the elastic plate, in December 1975. An origin in 1979 would treat the stresses imposed by the intrusions of the preceding four years as if their relaxation had not yet started. Because the opening was in fact spread over the episode, $t_0 = 1975$ gives the longest elapsed time compatible with the record, and the times since intrusion used below are upper bounds. The two epoch pairs then sit at $\tau = 12$–15 yr and 12–17 yr after the intrusion.

With this common origin, we fit both profiles jointly, because they record the same episode. A single (a, b) is shared by the two profiles, each is evaluated at its own (τ_i, τ_f), and the pooled χ^2 (48) is minimised over the $n = 37$ stations combined, with c as a third free parameter. The recurrence interval follows from $T = b/V$ with $V = 18$ mm yr^{-1}, the NUVEL-1A full spreading rate for north Iceland (DeMets et al. 1994), which Hofton & Foulger (1996) adopt as the time-averaged opening rate across the region. Foulger et al. (1992) used the 19 mm yr^{-1} predicted by the earlier NUVEL-1 model, which the NUVEL-1A revision of the geomagnetic time scale superseded.

Both models reproduce the near-rift step and the far-field plateau of the two profiles with $\chi^2_r < 2$ (Fig. 10a,b;

Table 1. Best-fit RSS parameters for the joint Krafla and Asal fits, with 1σ ranges in brackets ($\Delta\chi^2 \leq 2.30$ region projected onto each axis of the (a, b) landscape; at Krafla, c is minimised at every grid point). The recurrence interval is derived as $T = b/V$ ($V = 18$ mm yr^{-1} at Krafla, 13 mm yr^{-1} at Asal) and τ_R from a through (47); c is the reference-frame offset of the Hofton & Foulger (1996) profile, and p the number of free parameters. At Asal, χ^2/n is 1.37 (EV) and 1.62 (EVE) on the 33 time-series points and 21.3 and 25.4 on the 15 profile points. The Diffusion row is the Fickian model of Foulger et al. (1992) refit under the same conditions as the RSS, with $\kappa = 2.70$ m^2 s^{-1} in place of a; its 1σ ranges were not computed. ΔAIC is relative to the EV RSS at the same site.

Site	Model	a (10^{-5} m s^{-1})	b (m)	T (yr)	τ_R (yr)	c (cm)	p	χ^2_r	ΔAIC
Krafla ($n = 37$)	EV RSS	4.99 [4.21, 5.82]	1.71 [1.60, 1.80]	95 [89, 100]	–	6.5	3	1.61	–
	EVE RSS	11.9 [10.8, 13.1]	1.60 [1.51, 1.69]	89 [84, 94]	3.6 [3.2, 3.9]	6.6	3	1.82	+7.1
	Diffusion	–	2.81	–	–	6.4	3	4.90	+111.7
Asal ($n = 48$)	EV RSS	1.00 [0.97, 1.04]	0.673 [0.666, 0.681]	52 [51, 52]	–	–	2	7.93	–
	EVE RSS	6.95 [6.77, 7.14]	0.642 [0.637, 0.648]	49 [49, 50]	6.1 [5.9, 6.2]	–	2	9.46	+70.3

Table 1), and the data favour the EV RSS at the same parameter count (ΔAIC = +7.1).

Both χ^2 landscapes have a single closed minimum within the 3σ contour (Fig. 10c,d), so the short observation window still bounds both a and b (Table 1). The two parameters are nearly uncorrelated in the EV landscape and positively correlated in the EVE landscape.

For comparison, we refit the diffusion model of Fig. 1 under the same conditions as the RSS: one (κ, b) shared by the two profiles, $t_0 = 1975$, and the offset c of the Hofton & Foulger (1996) profile solved during the fit. The refit gives $\kappa = 2.70$ m^2 s^{-1}, $b = 2.81$ m and $c = 6.4$ cm, with $\chi^2_r = 4.90$ against 1.61 for the EV RSS at the same parameter count (ΔAIC = +111.7; Table 1).

4.2 Asal rift, Djibouti

The November 1978 Asal–Ghoubbet rifting episode in the Afar Triangle produced about 2 m of rift-perpendicular opening over several days (Le Dain et al. 1979; Ruegg et al. 1984). Vigny et al. (2007) tracked the EP00–DF00 baseline, a rift-perpendicular geodetic tie crossing the Asal rift, from 1979 to 2003. The baseline lengthened by ≈ 0.5 m, rapidly at first and much more slowly after the mid-1980s. Vigny et al. (2007) also measured rift-perpendicular GPS velocities along a transect crossing the rift, at a mean epoch of ≈ 19 yr after the event. We digitised all 33 time-series points from their Figure 3 and all 15 profile points from their velocity transect, and fit both jointly with the EV and EVE RSS forward models of Sections 2.5 and 3.3.

The predicted baseline change is $\Delta L = 2u(x_0, \tau)$ at $x_0 = 2$ km, half the EP00–DF00 rift-perpendicular separation, with the rifting onset fixed at $t_0 = 1978.85$ yr (Vigny et al. 2007). The predicted velocity profile is evaluated at the transect epoch, with the model centre placed at $x = +1$ km northeast of the topographic rift axis, following Vigny et al. (2007). The time series of Vigny et al. (2007) is referenced to the first measurement epoch, $\tau_{\rm ref} \approx 1.1$ yr after the rifting event, rather than to the onset. We therefore subtract from each model curve its own prediction at $\tau_{\rm ref}$, which anchors $\Delta L = 0$ at the reference epoch without introducing a free parameter.

As at Krafla, the recurrence interval follows from the opening-rate relation $T = b/V$. We take $V = 13$ mm yr^{-1}, the long-term Arabia–Somalia rate reported by Vigny et al. (2007). We do not use the ≈ 16 mm yr^{-1} opening they measure across the Asal rift itself: they attribute its excess over the long-term rate to the post-1978 transient, which the RSS already produces, so adopting it as V would count the transient twice.

The velocity profile is expressed in the Somalia reference frame, in which the SW flank is nearly fixed and the NE flank moves at the full rate. Because the model's far-field asymptotes are $\pm V/2$, the frame shift is $V/2$ and is not fitted. No second intrusion has been documented at Asal–Ghoubbet since 1978: Smittarello et al. (2016) analyse over 40 yr of geodetic monitoring (1972–2016) as the relaxation of that single event. We therefore require $T \geq T_{\rm min} = 37.2$ yr, the 1978–2016 span, which under $T = b/V$ becomes the lower bound $b \geq V\,T_{\rm min} \approx 0.48$ m. Only (a, b) remain free.

The EV RSS fits the Asal record better than the EVE RSS at equal parameter count (ΔAIC = +70.3; Fig. 11, Table 1). Both models reproduce the time series closely, with $\chi^2/n = 1.37$ for the EV RSS and 1.62 for the EVE RSS over its 33 points. The elevated joint χ^2_r comes from the profile ($\chi^2/n = 21.3$ and 25.4 over its 15 points), where the observed SW–NE asymmetry exceeds what a single symmetric transient can produce. The EVE profile also shows a velocity jump of ≈ 4.6 mm yr^{-1} at the model centre (Fig. 11b), the RSS jump of eq. (43) at $\tau \approx 19$ yr. Neither fit sits on the $T_{\rm min}$ floor.

Figure 11c,d shows the joint (time series + profile) χ^2 landscapes for both rheologies. The combined dataset confines (a, b) to a narrow, anti-correlated ellipse in both cases, well above the $T_{\rm min}$ floor, with 1σ ranges of a few per cent (Table 1).

5 DISCUSSION

5.1 Inter-rifting deformation from discrete intrusions

Kinematic models of inter-rifting deformation need a free parameter that the mechanics of a magmatic rift does not supply. Backslip models (Savage & Prescott 1978) and their dislocation equivalents (LaFemina et al. 2005; Konca et al. 2018) reproduce observed velocity gradients through a locking depth or a creep-zone geometry, which carries no independent constraint where the rheological transition is gradational. Numerical visco-elastic simulations meet the same obstacle at their lateral boundaries, which must either force the far-field velocity to zero, in contradiction with global

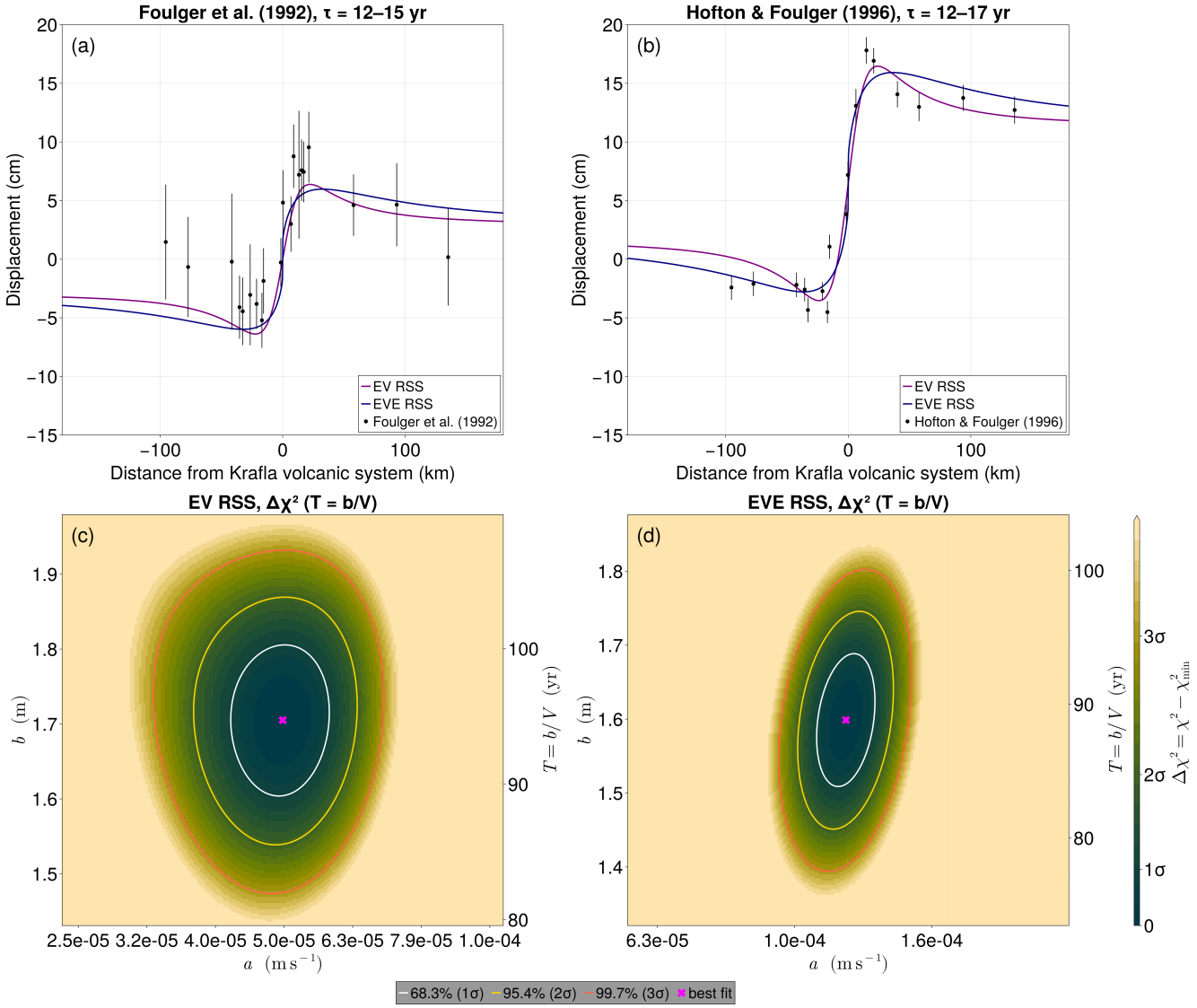


Figure 10. Joint EV and EVE RSS fits to both Krafla datasets, with T derived from the fitted dike opening b through the opening-rate relation $T = b/V$ at $V = 18$ mm yr^{-1} and time measured from $t_0 = 1975$. (a) Foulger et al. (1992) ($\tau_i = 12$ yr, $\tau_f = 15$ yr). (b) Hofton & Foulger (1996) ($\tau_i = 12$ yr, $\tau_f = 17$ yr), plotted at its own uncorrected displacements, with the reference-frame offset absorbed into each curve rather than removed from the data. Curves: EV RSS (purple) and EVE RSS (navy), each with one (a, b) shared by both panels; error bars are 1σ. (c, d) Joint χ^2 landscapes in (a, b) for the EV and EVE RSS, shown as $\Delta\chi^2 = \chi^2 - \chi^2_{\min}$, with the reference-frame offset minimised at every grid point. Contours mark the 68.3, 95.4 and 99.7 per cent confidence levels for two parameters ($\Delta\chi^2 = 2.30$, 6.17, 11.83); the cross marks the best fit. The right axis gives the derived $T = b/V$. See Table 1 for goodness-of-fit statistics.

plate motion models (Altamimi et al. 2016), or prescribe the divergence rate that the model is meant to predict (Hofton et al. 1995; Hofton & Foulger 1996).

The RSS removes both requirements because it treats inter-rifting deformation as the superposed relaxation of all past intrusions. Far-field plate divergence follows from the conservation of the opening over each cycle (eq. 25), and the only length scale is the deformation half-width $aT = EHT/[2\eta(1-\nu^2)]$, set by the balance between elastic stresses in the plate and viscous drag at its base. In the RSS, the steady velocity recorded far from the rift is the summed relaxation of intrusions that occurred many cycles earlier. A steady far-field velocity therefore cannot, on its own, be taken as evidence of creep at depth.

This dependence on past intrusions limits where the RSS applies. Because the step response decays only as $1/x$ (Section 2.4), the number of intrusions needed to reach the RSS grows linearly with distance from the axis, $N \gtrsim 13\,|x|/(aT)$ (Section 2.5). The farthest stations used here lie at $0.9\,aT$ at Krafla and $0.6\,aT$ at Asal. The RSS describes these stations provided Krafla and Asal have hosted more than about 12 and 8 intrusions, respectively, or about 1100 and 400 yr of rifting at the derived recurrence intervals.

5.2 Non-local versus Fickian relaxation

The EV model and the diffusion model of Foulger et al. (1992) differ in the order of their governing equation. The

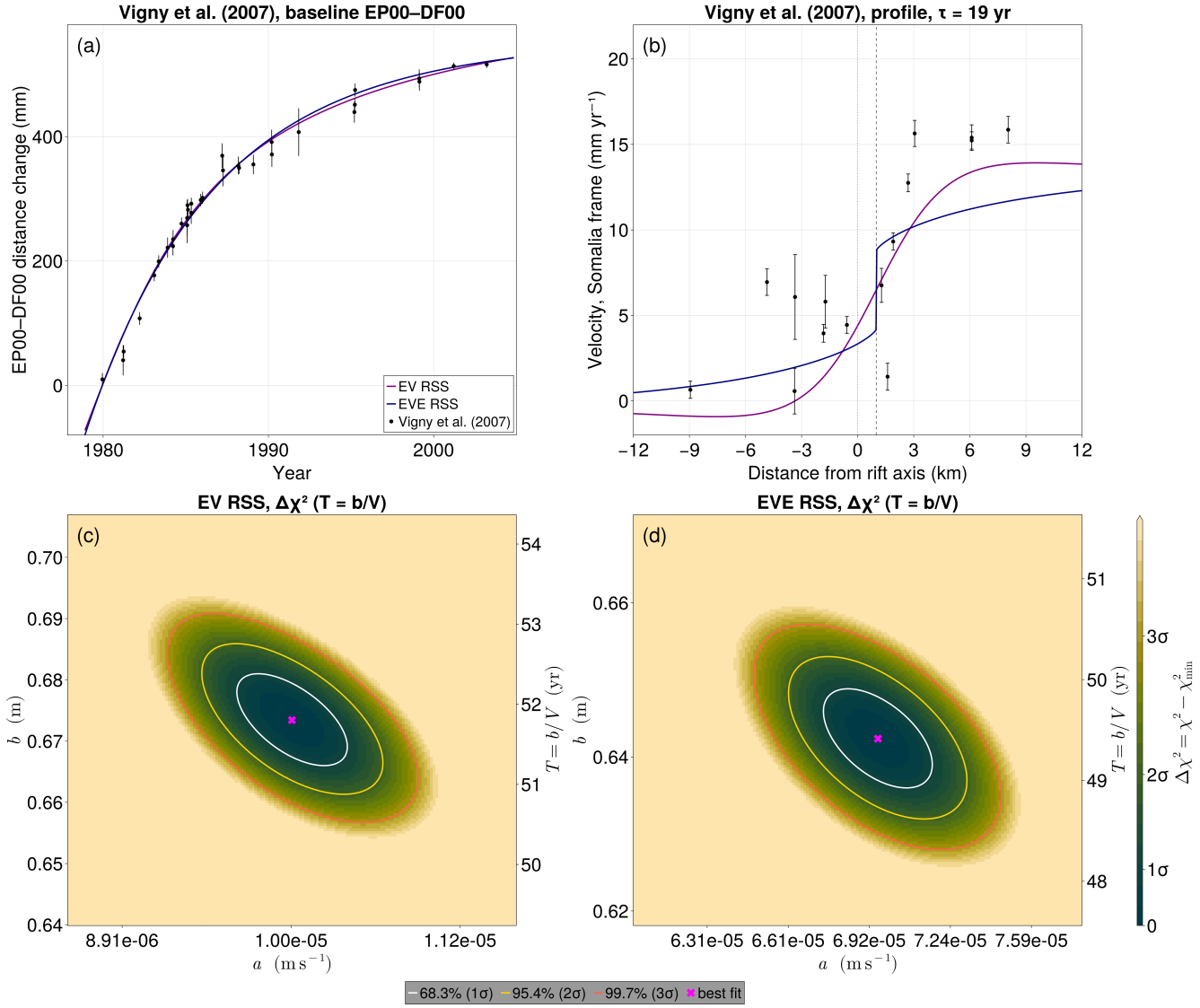


Figure 11. Joint EV and EVE RSS fits to the Asal data of Vigny et al. (2007), with T derived from the fitted dike opening b through the opening-rate relation $T = b/V$ at $V = 13$ mm yr^{-1}. (a) EP00–DF00 baseline change (mm), filled circles with 1σ error bars; model curves are referenced to the first measurement epoch ($\tau_{\rm ref} \approx 1.1$ yr). (b) Rift-perpendicular velocity profile at $\tau \approx 19$ yr in the Somalia reference frame (mm yr^{-1}); model curves include the frame shift $V/2$. The dotted line marks the topographic rift axis and the dashed line the model centre at $x = +1$ km. Curves: EV RSS (purple) and EVE RSS (navy), each with one (a, b) shared by both panels. (c, d) Joint χ^2 landscapes in (a, b) for the EV and EVE RSS, shown as $\Delta\chi^2 = \chi^2 - \chi^2_{\rm min}$. Contours mark the 68.3, 95.4 and 99.7 per cent confidence levels for two parameters ($\Delta\chi^2 = 2.30$, 6.17, 11.83); the cross marks the best fit. The right axis gives the derived $T = b/V$. See Table 1 for goodness-of-fit statistics.

Fickian model, $\partial u/\partial t = \kappa\,\partial^2 u/\partial x^2$, damps each Fourier mode at the rate κk^2; the EV model (eq. 8) damps it at $a|k|$, the Fourier signature of the half-Laplacian. This operator arises from the $1/(x-x')^2$ Stokes kernel (3), through which the half-space transmits stress instantaneously, so that every point of the plate starts to move as soon as the dike opens. As a result, the step velocity is a Lorentzian with a $1/x$ tail (18), against the Gaussian decay of the Fickian velocity (19, Fig. 4b).

Despite this difference, the EV decay factor equals an average of Fickian decay factors over an effective diffusion time s, a relation known as Bochner subordination (Metzler & Klafter 2000),

$$e^{-a|k|t} = \int_0^\infty e^{-k^2 s}\, g_t(s)\, ds, \qquad g_t(s) = \frac{at}{2\sqrt{\pi}}\, s^{-3/2} \exp\!\left(-\frac{a^2 t^2}{4s}\right), \tag{49}$$

so the EV step displacement (14) equals the Fickian profile $(b/2)\,\mathrm{erfc}[x/(2\sqrt{s})]$ averaged over s with the weight $g_t(s)$. We checked this identity numerically to a relative precision better than 10^{-7}. The weight peaks at $s = (at)^2/6$ but decays only as $s^{-3/2}$, so long effective diffusion times contribute substantially, and these carry the $1/x$ tail. The Fickian model retains a single diffusion time, $s = \kappa t$, and discards that tail.

The two formulations leave distinct, observable signa-

tures. The EV velocity peak sits at $x = at$ and migrates outward at the constant speed a, whereas the Fickian peak sits at $x = \sqrt{2\kappa t}$ and decelerates as $t^{-1/2}$ (Fig. 4b). The peak passes a station at distance x at $t = x/a$ in the EV model but at $t = x^2/(2\kappa)$ in the Fickian model, so its arrival time grows linearly with distance in the first case and quadratically in the second. The velocity recorded at that station also evolves differently. In the EV model it decreases monotonically from the intrusion, staying nearly constant at $ba/(\pi x)$ until $t \approx x/a$ and then falling as $1/t^2$. In the Fickian model it rises from zero to a maximum at $t = x^2/(6\kappa)$ before decaying. In the far field, the RSS settles onto the plate velocity $\pm V/2$, while the Fickian profile returns to its reference level. At Krafla, the diffusion model refit under the same conditions as the RSS (Section 4.1) fits the two profiles with $\chi^2_r = 4.90$, against 1.61 for the EV RSS. Its curves peak ≈ 50 km from the axis, whereas the data peak at 15–21 km, and they return to their reference level within ~ 150 km. Networks that span both the near-rift peak and the far-field plateau, or time series at several distances from the axis, are best placed to discriminate between the two formulations.

5.3 Distinguishing a viscous from a visco-elastic half-space

At equal a, the EVE model departs from the EV model only at short wavelengths and early times. The elastic response of the Maxwell half-space saturates the decay rate of modes with $k \gtrsim 1/\ell_M$, where ℓ_M is the Maxwell length (eq. 35), and its effect fades after a few τ_R. Its signature is a finite velocity at the axis that decays as e^{-t/τ_R} (eq. 36), which in the RSS becomes a velocity jump across the axis (eq. 43). This jump follows from treating the dike as infinitely thin and the plate as a single layer; in reality we expect it to be spread over a distance comparable to the plate thickness H. Under the shared-modulus closure, $\ell_M = H/(1-\nu)$ (eq. 47), about 13 km for $H = 10$ km, independently of a, so the elastic signal is confined within a distance of about $1.3\,H$ from the axis.

Whether a record can distinguish the two rheologies therefore depends on its first epoch in units of τ_R and on the fraction of its stations that lie within ℓ_M. At Krafla, the observations start $3.4\,\tau_R$ after the onset, and only five of 20 and three of 17 stations lie within $\ell_M \approx 13$ km of the axis. Most of the elastic contribution has decayed by then, and the data favour the EV RSS only modestly ($\Delta\mathrm{AIC} = +7.1$). At Asal, the time series starts at $0.2\,\tau_R$ and the whole velocity profile lies within ℓ_M. This record samples the window where the two rheologies differ most, and the data favour the EV RSS strongly ($\Delta\mathrm{AIC} = +70.3$).

This preference rests on the shared-modulus closure, which ties τ_R and ℓ_M to a and H; a half-space shear modulus different from that of the plate would shift both scales. The choice of rheology also matters for the inferred viscosity. The EVE fits require a larger characteristic velocity than the EV fits, by a factor of 2.4 at Krafla and 6.9 at Asal (Table 1), which lowers η to 1.8×10^{18} and 3.1×10^{18} Pa s at $H = 10$ km and $E = 40$ GPa. At Asal, this rheological uncertainty exceeds the χ^2 uncertainty on a by more than an order of magnitude. Viscosities inferred from post-diking geodesy are therefore conditional on the assumed half-space rheology as much as on the elastic parameters.

5.4 Inferred viscosity and model limitations

The fits constrain the composite parameter $a = EH/[2\eta(1-\nu^2)]$. For fixed E, the best-fit a maps onto an iso-a band in the (H, η) plane (Figs 12 and 13). An independent estimate of H, for which we take the seismogenic thickness, $H \approx 5$–15 km in Iceland and the Afar (Stevens et al. 2021; Plasman et al. 2017; Weinstein et al. 2017), turns it into a viscosity. For the canonical $H = 10$ km, $E = 40$ GPa and $\nu = 0.25$, the EV RSS gives $\eta \approx 4.3 \times 10^{18}$ Pa s at Krafla and $\eta \approx 2.1 \times 10^{19}$ Pa s at Asal. The 1σ ranges from the χ^2 landscapes alone are narrow, $[3.7, 5.1] \times 10^{18}$ and $[2.05, 2.21] \times 10^{19}$ Pa s. Adding an uncertainty on E over [20, 80] GPa in quadrature widens them to $[2.1, 8.7] \times 10^{18}$ and $[1.1, 4.3] \times 10^{19}$ Pa s, so the elastic modulus dominates the error budget. The Krafla value lies below the bound $\eta < 10^{19}$ Pa s of Sigmundsson (1991) and within the post-glacial rebound range for Iceland (Sigmundsson et al. 2010), consistent with the hot sub-lithospheric mantle of the Iceland plume (Holtzman 2016). The Asal value is five times higher, and higher than estimates for the Dabbahu segment of the same rift system. There, Nooner et al. (2009) find $\eta \approx 5 \times 10^{18}$ Pa s beneath a 12–14 km elastic layer, and Hamling et al. (2014) find $\eta \approx 10^{18}$–$10^{18.5}$ Pa s beneath a $\sim$21 km elastic layer once magmatic sources are included. At Asal itself, a thermo-mechanical model of the first 6–8 yr after rifting gives a near-axis effective viscosity of 0.5–1×10^{18} Pa s (Cattin et al. 2005, Fig. 13), well below our estimate. This is consistent with a near-axis viscosity lower than the laterally averaged value that the 25-yr geodetic record constrains.

The deformation half-width explains why the profiles constrain b at all (Fig. 14). With the derived recurrence intervals, $aT \approx 149$ km at Krafla and ≈ 16 km at Asal, comparable to the extent of each network on either side of the rift. Each network therefore samples the transition from the near-rift transient to the far-field plateau $\pm V/2$, and through $T = b/V$ this transition fixes the opening.

Several misfits remain. At Krafla, the derived $T = 89$–95 yr is about a third of the ~ 250 yr between the 1724–1729 and 1975–1984 crises. At Asal, the fitted opening $b = 0.67$ m is about a third of the ≈ 1.9 m measured across the rift after the 1978 event (Ruegg et al. 1979). The predicted early lengthening rate exceeds the trilateration rate by 27 per cent (Ruegg & Kasser 1987), and the profile retains an SW–NE asymmetry that a symmetric model cannot produce ($\chi^2/n \approx 21$). Each misfit points to a simplification of the model, which assumes a single, infinitely thin dike in a single elastic plate, symmetry about the axis, strictly periodic recurrence and no shallow magmatic source. Smittarello et al. (2016) argue that magma supply rather than visco-elastic relaxation controls the post-diking velocity field at Asal. At Dabbahu, Hamling et al. (2014) likewise find that relaxation alone cannot reproduce the post-rifting deformation, and that inflation and deflation of magma bodies within the rift segment account for a large part of the signal. Our model contains relaxation only, so any contribution from magma supply is absorbed into the fitted (a, b); separating the two requires an independent constraint on the magmatic budget. The fitted parameters are therefore effective values for a symmetric, periodic rift, and the viscosities derived from them inherit this simplification.

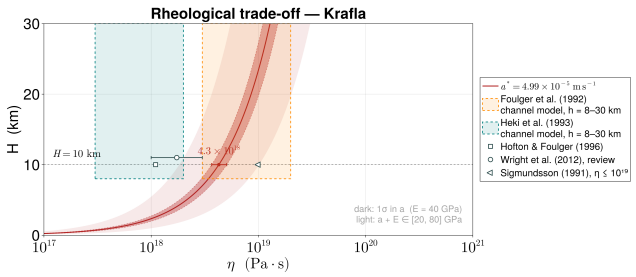


Figure 12. Rheological trade-off for Krafla, for the EV RSS best-fit $a = 4.99 \times 10^{-5}$ m s^{-1} (Section 4.1). The panel shows the iso-a contour in the (H, η) plane at the best-fit value. Dark shading propagates the 1σ uncertainty on a from the joint χ^2 landscape at fixed $E = 40$ GPa; light shading additionally propagates E over [20, 80] GPa (combined 1σ envelope). The $H = 10$ km dashed line marks the reference plate thickness, and the filled circle with error bar the best fit at that thickness with its 1σ range at $E = 40$ GPa. Published estimates: dashed boxes, the channel-model viscosities of Foulger et al. (1992) and Heki et al. (1993) for an elastic layer of 8–30 km (thin-channel viscosities, not directly comparable); open square, Hofton & Foulger (1996); open circle with error bar, Wright et al. (2012), drawn at $H = 11$ km for legibility; open triangle, the upper bound $\eta \leq 10^{19}$ Pa s of Sigmundsson (1991).

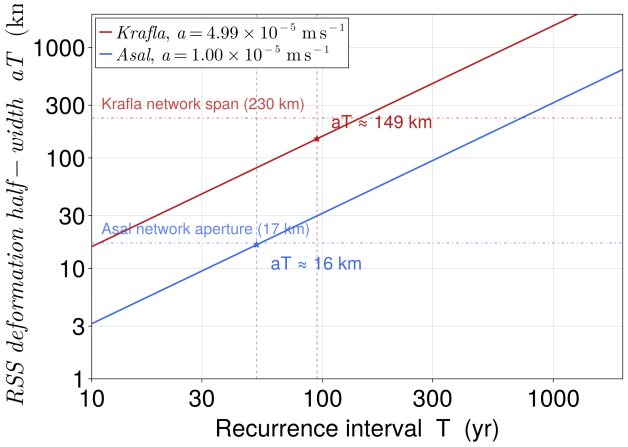


Figure 14. RSS deformation half-width aT as a function of recurrence interval T for Krafla (red, $a = 4.99 \times 10^{-5}$ m s^{-1}) and Asal (blue, $a = 1.00 \times 10^{-5}$ m s^{-1}), the EV RSS best fits. Vertical dashed lines mark the recurrence intervals derived from $T = b/V$ (95 yr and 52 yr); stars give the implied aT. Horizontal dash-dot lines mark the total span of each geodetic network used in this study: 230 km for Krafla (distance between outermost stations; Foulger et al. 1992) and 17 km for Asal (GPS profile extent; Vigny et al. 2007).

5.5 Irregular recurrence

The RSS assumes that intrusions recur at a fixed interval T, with a fixed opening b and at a fixed position. Natural rifts depart from all three. The $\sim$ 250 yr gap between the Krafla crises is a single estimate from a sparse historical record, dike openings vary within and between sequences (Thordarson & Larsen 2007), and successive dikes need not open on the same plane. To assess these departures, we summed the EV step velocities (15) of many past intrusions whose timing, opening and position vary from cycle to cycle, keeping $T = b/V$ on average (Appendix E). We considered four cases (Fig. 15a–d). In the first, openings b_j vary at random while the interval stays fixed, so that opening and interval are uncoupled. In the second, intervals T_j vary and each opening matches the preceding interval, $b_j = VT_j$. In the third, timing and opening are periodic but the dike position x_j scatters about the axis with standard deviation $\sigma_x = 0.1\,aT$. The fourth combines the second and third.

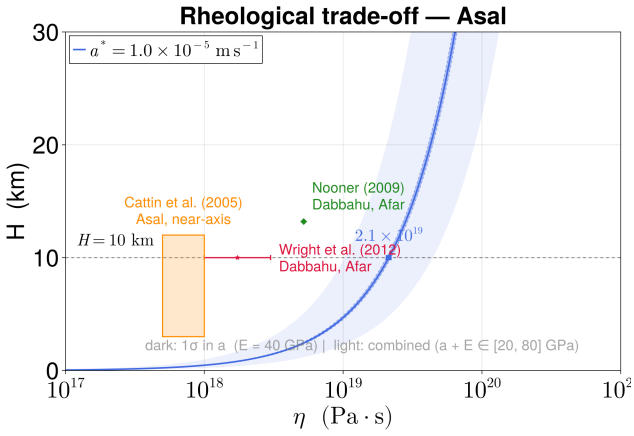


Figure 13. Rheological trade-off for Asal, for the EV RSS best-fit $a = 1.00 \times 10^{-5}$ m s^{-1} (Section 4.2). The panel shows the iso-a contour in the (H, η) plane at the best-fit value. Dark shading propagates the 1σ uncertainty on a from the joint χ^2 landscape at fixed $E = 40$ GPa; light shading additionally propagates E over [20, 80] GPa (combined 1σ envelope). The $H = 10$ km dashed line marks the reference plate thickness; markers show published viscosity estimates: filled diamond, Nooner et al. 2009, and star with error bar, Wright et al. 2012, both for the Dabbahu segment of the same rift system; orange box, Cattin et al. 2005: near-axis effective viscosity $\eta = 0.5$–1×10^{18} Pa s from a 2D thermo-mechanical finite-element model of the first 6–8 yr after rifting at Asal, with $H = 3$–12 km from the same study.

In all four cases the ensemble median follows the deterministic RSS to within 3 per cent of the far-field velocity $V/2$. For scatter in b_j or x_j alone, linearity fixes the ensemble mean: it equals the RSS for b_j, and the RSS convolved with the distribution of x_j for x_j. Scatter in T_j enters non-linearly through the ages of past intrusions, but the resulting bias is small at $c_v = 0.30$. The RSS therefore describes the typical velocity field of an irregular rift, and the bands measure how far a single realisation departs from it.

The three sources of scatter act at different distances from the axis, because each station integrates a different set of past intrusions. Near the axis, the contribution of an intrusion decays with its age t as $1/t^2$ (eq. 15), so the most recent ones dominate; farther out, a station also records the contribution of every intrusion younger than about $|x|/a$. In units of $V/2$, the 10–90 per cent band of the uncoupled case (a) reaches 0.46 at $x = 0.3\,aT$ and still 0.20 at $x = 5\,aT$. When openings do not match the intervals, the plateau of a single realisation reflects the opening rate realised over the last few cycles. Coupling the openings to the intervals (b) removes most of that far-field spread (0.31 at $x = 0.3\,aT$, 0.06 at $2\,aT$, 0.01 at $5\,aT$), because the opening delivered over any time window then equals V times that window. Scatter in dike position (c) acts only near the axis, where the band reaches 0.62 of $V/2$, and decays to 0.24 at $x = 0.3\,aT$

and below 0.01 at $x = 5\,aT$. Far from the axis, a shift of the source by σ_x barely changes the distance to the station. Combined, coupled timing and dike position (d) give a band of 0.6 at the axis that narrows to 0.18 at $x = aT$ and 0.02 at $x = 5\,aT$.

These results bear directly on the fits of Section 4. Both networks lie within one half-width of the axis, $|x| \lesssim 0.9\,aT$ at Krafla and $|x| \lesssim 0.6\,aT$ at Asal, where the bands of the coupled and dike-position cases are widest. The effect of dike position scales with $\sigma_x/(aT)$, so it matters more at Asal than at Krafla: a dike zone a few kilometres wide corresponds to $\sigma_x/(aT)$ of order 0.1 at Asal ($aT \approx 16$ km) but of order 0.01 at Krafla ($aT \approx 149$ km). The fitted (a, b) therefore carry an uncertainty from cycle-to-cycle variability that the χ^2 landscapes do not include. The bands of Fig. 15 are computed at one phase and one level of scatter, so they indicate the size of this effect rather than quantify it for either site.

5.6 Perspectives

The RSS velocity depends on three combinations of the model parameters: the half-width aT, the far-field velocity $b/(2T) = V/2$, and the phase τ/T (eq. 23). Our fits derive the recurrence interval from the fitted opening and an independent rift opening rate, $T = b/V$ (Section 4). Records that span a larger fraction of a cycle, or snapshots repeated over decades, would sample the evolution of the profile with phase and so test this derived T against geodesy. Propagating the stochastic spread of Section 5.5 into the inferred parameters, with ensembles conditioned on each site's phase and intrusion history, would complete the uncertainty budget.

Real rifts are segmented: dikes span finite along-strike lengths, and segments differ in recurrence history, plate thickness and heat flow. Because the only length scale of the model is the local product aT, each segment generates its own deformation halo, and to the extent that the segments respond linearly, the surface velocity is their superposition. The 1D model cannot capture along-strike dike propagation or the interaction between segments. Extending the RSS to the full 2D problem requires relaxing the plane-strain approximation while retaining the non-local Stokes coupling at the base of the plate.

6 CONCLUSIONS

Steady inter-rifting velocities at magmatic rifts can arise from discrete dike intrusions alone. In the RSS, the relaxation of past intrusions produces the far-field plate divergence, and the deformation half-width aT is the only length scale. Because old intrusions carry this divergence, the RSS describes a station at distance $|x|$ only if the rift has hosted more than about $13\,|x|/(aT)$ intrusions. On young rift segments, or far beyond aT, the finite sum of past intrusions should replace it.

Above a viscous half-space, post-diking relaxation is governed by the half-Laplacian, so the velocity peak migrates at the constant speed a and the profile decays as $1/x$. The EV solution averages Fickian profiles over a heavy-tailed distribution of diffusion times. The Fickian model of

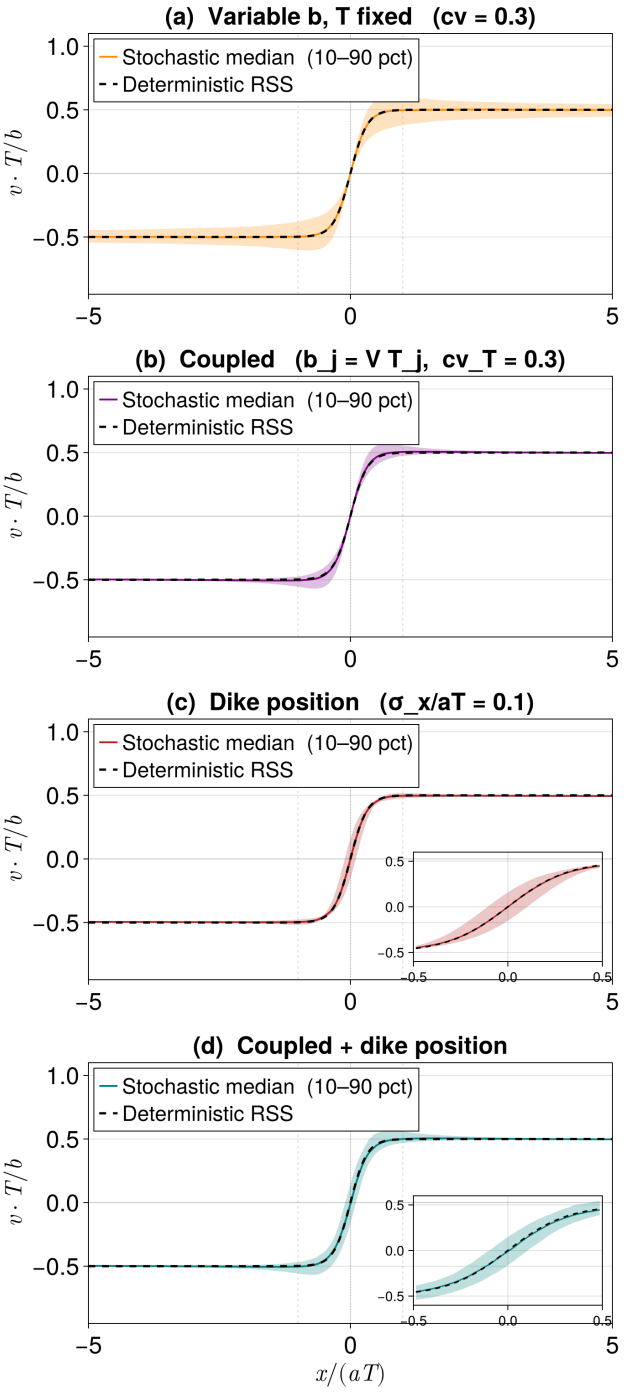


Figure 15. EV RSS velocity profile under stochastic recurrence ($a = T = b = 1$; phase $\tau/T = 0.5$; 300 past intrusions; 500 realisations; $T = b/V$ on average in every case). Velocity is normalised by b/T and distance by aT. Shaded bands show the 10th–90th percentile range; coloured lines are ensemble medians; the black dashed line is the deterministic RSS (23); vertical dashed lines mark $x = \pm aT$. (a) Uncoupled: openings b_j random ($c_v = 0.30$) with T fixed. (b) Intervals T_j random ($c_v = 0.30$) with each opening coupled to the preceding interval, $b_j = VT_j$. (c) Dike position x_j random, Normal with $\sigma_x = 0.1\,aT$, periodic timing and opening. (d) Cases (b) and (c) combined. Insets in (c) and (d) enlarge $|x| \le 0.5\,aT$.

Foulger et al. (1992) retains a single diffusion time and therefore lacks the $1/x$ tail. Under the same fitting conditions at Krafla, the EV RSS fits the two post-rifting profiles with $\chi^2_r = 1.61$, against 4.90 for the diffusion model.

A geodetic record can distinguish a Maxwell half-space from a viscous one only if it starts within a few relaxation times τ_R of the intrusion and includes stations within $\ell_M = H/(1-\nu) \approx 1.3\,H$ of the axis. The Asal record meets both conditions and strongly favours the viscous half-space (ΔAIC $= +70.3$). The Krafla record, which starts $3.4\,\tau_R$ after the onset, favours it only modestly (ΔAIC $= +7.1$). The two rheologies imply viscosities that differ by up to a factor of seven, so viscosities inferred from post-diking geodesy depend on the assumed half-space rheology.

Tying T to an independent opening rate leaves two parameters, a and b, which both records constrain. For $H = 10$ km and $E = 40$ GPa, the EV RSS implies $\eta \approx 4.3 \times 10^{18}$ Pa s at Krafla and 2.1×10^{19} Pa s at Asal, and the uncertainty on E dominates the error budget. The remaining misfits reflect the simplified source geometry and strictly periodic recurrence of the model, so the fitted parameters are effective values for a symmetric, periodic rift.

Irregular recurrence leaves the RSS largely unbiased. With 30 per cent scatter in interval and opening, and dike positions scattered over $0.1\,aT$, the ensemble median stays within 3 per cent of the deterministic RSS. Individual profiles scatter most within one half-width of the axis, where both networks lie, so cycle-to-cycle variability adds uncertainty to the fitted (a, b) beyond the χ^2 landscapes.

ACKNOWLEDGEMENTS

[PLACEHOLDER — funding agencies, field/computing support, reviewer thanks.]

Data Availability

The geodetic data used in this study are from published sources cited in the text. The analytical model code and inversion scripts are available at [PLACEHOLDER — repository DOI or URL upon acceptance].

REFERENCES

Ablowitz, M. J. & Fokas, A. S., 2003. *Complex Variables: Introduction and Applications*, Cambridge University Press, Cambridge, UK, 2nd edition.

Ali, S. T. et al., 2014. Geodetic measurements of postseismic deformation following the 2005 Kashmir (Pakistan) earthquake, *Geophysical Journal International*.

Altamimi, Z. et al., 2016. ITRF2014: A new release of the International Terrestrial Reference Frame, *Journal of Geophysical Research*.

Barbot, S. & Fialko, Y., 2010. A unified continuum representation of post-seismic relaxation mechanisms and the link with precession, *Geophysical Journal International*, **182**, 1124–1140.

Bürgmann, R. & Dresen, G., 2008. Rheology of the lower crust and upper mantle: evidence from rock mechanics, geodesy, and field observations, *Annual Review of Earth and Planetary Sciences*, **36**, 531–567.

Cattin, R., Doubre, C., de Chabalier, J.-B., King, G., Vigny, C., Avouac, J.-P., & Ruegg, J.-C., 2005. Numerical modelling of quaternary deformation and post-rifting displacement in the Asal–Ghoubbet rift (Djibouti, Africa), *Earth and Planetary Science Letters*, **239**, 352–367, doi: 10.1016/j.epsl.2005.07.028.

Cohen, S. C. & Kramer, M. J., 1984. Crustal deformation, the earthquake cycle, and models of viscoelastic flow, *Geophysical Journal International*.

Corti, G., 2009. Continental rift evolution: From rift initiation to incipient break-up, *Earth-Science Reviews*.

DeMets, C., Gordon, R. G., Argus, D. F., & Stein, S., 1994. Effect of recent revisions to the geomagnetic reversal time scale on estimates of current plate motions, *Geophysical Research Letters*, **21**(20), 2191–2194.

Ebinger, C. J. et al., 2013. Rifting in diverse tectonic settings, *Tectonophysics*.

Fernandes, R. M. S. et al., 2004. Defining the plate boundaries in the Azores region, *Journal of Volcanology and Geothermal Research*.

Foulger, G. R. et al., 1992. Post-rifting stress relaxation at the divergent plate boundary in NE Iceland, *Nature*, **358**, 488–490.

Grandin, R. et al., 2011. Seismicity during lateral dike propagation, *Geophysical Research Letters*.

Hamling, I. J., Wright, T. J., Calais, E., Lewi, E., & Fukahata, Y., 2014. InSAR observations of post-rifting deformation around the Dabbahu rift segment, Afar, Ethiopia, *Geophysical Journal International*, **197**(1), 33–49, doi: 10.1093/gji/ggu003.

Heki, K. et al., 1993. Plate dynamics near divergent boundaries: Geophysical implications of postrifting crustal deformation in NE Iceland, *Journal of Geophysical Research*, **98**, 14279–14297.

Hetland, E. A., 2006. *Postseismic relaxation and interseismic coupling*, Ph.D. thesis, Massachusetts Institute of Technology.

Hetland, E. A. & Hager, B. H., 2005. Postseismic and interseismic displacements near a strike-slip fault: A two-dimensional theory for general linear viscoelastic rheologies, *Journal of Geophysical Research*, **110**, B10401.

Hetland, E. A. & Hager, B. H., 2006. Interseismic strain accumulation: Spin-up, cycle invariance, and irregular rupture sequences, *Geochemistry, Geophysics, Geosystems*, **7**, Q03010.

Hofton, M. A. et al., 1995. Horizontal surface deformation in Iceland 1993, *Journal of Geophysical Research*.

Hofton, M. A. & Foulger, G. R., 1996. Post-rifting anelastic deformation around the spreading plate boundary, north Iceland, *Journal of Geophysical Research*.

Holtzman, B. K., 2016. Questions on the existence, persistence, and mechanical effects of a very small melt fraction in the asthenosphere, *Geochemistry, Geophysics, Geosystems*.

Konca, A. O. et al., 2018. Slip distribution and moment tensor of the 2010 Mw 7.2 Baja California earthquake, *Geophysical Journal International*.

Kreemer, C. et al., 2014. A geodetic plate motion and global strain rate model, *Geochemistry, Geophysics, Geosystems*.

LaFemina, P. C. et al., 2005. Geodetic GPS measurements in south Iceland: Strain accumulation and partitioning in a propagating ridge system, *Journal of Geophysical Research*.

Laurie, D. P., 1997. Calculation of Gauss–Kronrod quadrature rules, *Mathematics of Computation*, **66**, 1133–1145.

Le Dain, A. Y. et al., 1979. Seismotectonic aspects of the Asal–Ghoubbet rift, *Bulletin de la Société Géologique de France*.

Maccaferri, F. et al., 2016. Time-dependent rift magma intrusions and their effect on seismicity, *Geophysical Research Letters*.

Metzler, R. & Klafter, J., 2000. The random walk's guide to anomalous diffusion, *Physics Reports*, **339**, 1–77.

Nooner, S. L. et al., 2009. Post-rifting relaxation in the Afar region, Ethiopia, *Geophysical Research Letters*, **36**, L21308.

Nur, A. & Mavko, G., 1974. Postseismic viscoelastic rebound,

Science, **183**, 204–206.
Plasman, M. et al., 2017. Lithospheric strength variations in Mainland China: Tectonic implications, *Geophysical Journal International*.
Prodehl, C. et al., 1997. The East African rift system — a review of crustal structure studies, *Tectonophysics*.
Reiss, M. C. et al., 2022. Shallow seismicity and the classification of structures in the Natron Basin, *Tectonophysics*.
Ruegg, J. C. et al., 1984. Seismotectonic aspects of the Asal rift system, *Tectonophysics*, **104**, 153–183.
Ruegg, J. C. & Kasser, M., 1987. Deformation across the Asal–Ghoubbet rift, Djibouti: uplift and crustal extension 1979–1986, *Geophysical Research Letters*, **14**, 745–748.
Ruegg, J. C., Lépine, J. C., Tarantola, A., & Kasser, M., 1979. Geodetic measurements of rifting associated with a seismo-volcanic crisis in Afar, *Geophysical Research Letters*, **6**, 817–820.
Samko, S. G. & Ross, B., 1993. Integration and differentiation to a variable fractional order, *Integral Transforms and Special Functions*, **1**, 277–300.
Samko, S. G., Kilbas, A. A., & Marichev, O. I., 1993. *Fractional Integrals and Derivatives: Theory and Applications*, Gordon and Breach, Yverdon.
Saria, E. et al., 2014. Present-day kinematics of the East African rift, *Journal of Geophysical Research*.
Savage, J. C. & Prescott, W. H., 1978. Asthenosphere readjustment and the earthquake cycle, *Journal of Geophysical Research*, **83**, 3369–3376.
Sigmundsson, F., 1991. Post-glacial rebound and asthenosphere viscosity in Iceland, *Geophysical Research Letters*, **18**(6), 1131–1134, doi: 10.1029/91GL01342.
Sigmundsson, F. et al., 2010. Geodetic constraints on the properties of the lithosphere in Iceland, *Journal of Geophysical Research*.
Smittarello, D., Grandin, R., De Chabalier, J.-B., Doubre, C., Deprez, A., Masson, F., Socquet, A., & Saad, I. A., 2016. Transient deformation in the Asal-Ghoubbet Rift (Djibouti) since the 1978 diking event: Is deformation controlled by magma supply rates?, *Journal of Geophysical Research: Solid Earth*, **121**(8), 6030–6052.
Spence, D. A. & Turcotte, D. L., 1979. Magma-driven propagation of cracks, *Journal of Geophysical Research*.
Stamps, D. S. et al., 2018. A virtual geodetic observatory for monitoring East African Rift deformation, *Geochemistry, Geophysics, Geosystems*.
Stamps, D. S. et al., 2021. Redefining East African Rift System kinematics, *Geology*.
Stevens, V. L., Sloan, R. A., Chindandali, P. R., Wedmore, L. N. J., Salomon, G. W., & Muir, R. A., 2021. The entire crust can be seismogenic: Evidence from southern Malawi, *Tectonics*, **40**(6), e2020TC006654, doi: 10.1029/2020TC006654.
Thatcher, W. & Rundle, J. B., 1984. A viscoelastic coupling model for the cyclic deformation due to periodically repeated earthquakes at subduction zones, *Journal of Geophysical Research*, **89**, 7631–7640.
Thordarson, T. & Larsen, G., 2007. Volcanism in Iceland in historical time: volcano types, eruption styles and eruptive history, *Journal of Geodynamics*, **43**, 118–152.
Vigny, C. et al., 2007. Twenty-five years of geodetic measurements along the Tadjoura–Asal rift system, Djibouti, East Africa, *Journal of Geophysical Research*.
Weinstein, A. et al., 2017. Fault-magma interactions during early continental rifting: Seismicity of the magadi-natron-manyara basins, africa., *Geochemistry, Geophysics, Geosystems*.
Wright, T. J. et al., 2012. Geophysical constraints on the dynamics of spreading centres from rifting episodes on land, *Nature Geoscience*, **5**, 242–250.

APPENDIX A: FOURIER TRANSFORM OF THE STEP INITIAL CONDITION

The Fourier transform of the initial condition (11) is $\hat{u}(k,0) = b\int_{-\infty}^{0} e^{-ikx}\,dx$, which for real k is treated in the distributional sense by adding a convergence regulator $\varepsilon > 0$:

$$\int_{-\infty}^{0} e^{-ikx}\,dx = \lim_{\varepsilon\to 0}\int_{-\infty}^{0} e^{(-ik+\varepsilon)x}\,dx = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon - ik}.$$

By the Sokhotski–Plemelj theorem (Ablowitz & Fokas 2003),

$$\lim_{\varepsilon\to 0^+}\frac{1}{x \pm i\varepsilon} = \mathcal{P}\left(\frac{1}{x}\right) \mp i\pi\delta(x), \tag{A.1}$$

where $\mathcal{P}$ denotes the Cauchy principal value. Multiplying numerator and denominator by i recasts the regulator as $+i\varepsilon$,

$$\frac{1}{\varepsilon - ik} = \frac{i}{i\varepsilon + k} = \frac{i}{k + i\varepsilon},$$

so that (A.1) applies directly with $x = k$ and the upper sign:

$$\begin{aligned}\lim_{\varepsilon\to 0^+}\frac{1}{\varepsilon - ik} &= \lim_{\varepsilon\to 0^+}\frac{i}{k+i\varepsilon}\\ &= i\left[\mathcal{P}\left(\frac{1}{k}\right) - i\pi\delta(k)\right]\\ &= \pi\delta(k) + \frac{i}{k},\end{aligned} \tag{A.2}$$

where the principal-value symbol on $1/k$ is dropped, as is standard, since it is understood implicitly whenever $1/k$ appears next to a $\delta(k)$ term. This gives (12).

The inverse transform of the evolved spectrum (13) splits into two contributions, $u(x,t) = I_1 + I_2$. The delta-function term gives a constant background $I_1 = b/2$; for I_2, the parity of $\sin(kx)/k$ yields

$$I_2 = -\frac{b}{\pi}\int_0^\infty \frac{\sin(kx)}{k}\,e^{-akt}\,dk = -\frac{b}{\pi}\arctan\!\left(\frac{x}{at}\right), \tag{A.3}$$

a standard Dirichlet-type integral. With the complementary arctangent identity $\arctan(z) + \arctan(1/z) = \pi/2$, the displacement counted from the instant after the intrusion is the step solution (14).

APPENDIX B: DIGAMMA REPRESENTATION OF THE RSS VELOCITY

Each term of the RSS series (22) splits into partial fractions,

$$\frac{x}{x^2 + a^2(\tau + jT)^2} = \frac{1}{2i}\left[\frac{1}{a(\tau+jT) - ix} - \frac{1}{a(\tau+jT) + ix}\right].$$

Factoring aT from each denominator and substituting into (22), the sum takes the form of a difference of two Digamma series. With the representation $\psi(z) = -\gamma + \sum_{j=0}^{\infty}[1/(j+1) - 1/(j+z)]$ and $z = \tau/T \pm ix/(aT)$, the Euler–Mascheroni constant γ and the $1/(j+1)$ terms cancel when the two conjugate series are differenced. With $Z = \tau/T + ix/(aT)$ and the identity $\psi(Z) - \psi(\bar{Z}) = 2i\,\mathrm{Im}\{\psi(Z)\}$, the series reduces to (23).

APPENDIX C: CORRESPONDENCE PRINCIPLE FOR THE MAXWELL HALF-SPACE

For deviatoric stress σ and strain ε, the 1D constitutive relation of a Maxwell body is

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{2\mu_m} + \frac{\sigma}{2\eta}. \tag{C.1}$$

Applying the Laplace transform $\tilde{f}(s) = \mathcal{L}\{f(t)\}$ to (C.1) with zero initial conditions gives

$$s\tilde{\varepsilon} = \tilde{\sigma}\left(\frac{s}{2\mu_m} + \frac{1}{2\eta}\right). \tag{C.2}$$

Solving for stress in terms of the strain rate $s\tilde{\varepsilon}$ gives $\tilde{\sigma} = 2\tilde{\eta}(s)\,(s\tilde{\varepsilon})$, with the effective viscosity $\tilde{\eta}(s)$ of (26). Laplace-transforming the Stokes kernel (3) and replacing η by $\tilde{\eta}(s)$ gives

$$\tilde{\sigma}_{xz}^{\mathrm{BOT}}(x,s) = \int_{-\infty}^{+\infty} \frac{2\tilde{\eta}(s)}{\pi(x-x')^2}\left[s\tilde{u}(x',s)\right] dx',$$

or, with (26),

$$(1+\tau_R s)\,\tilde{\sigma}_{xz}^{\mathrm{BOT}}(x,s) = \int_{-\infty}^{+\infty} \frac{2\eta}{\pi(x-x')^2}\left[s\tilde{u}(x',s)\right] dx'. \tag{C.3}$$

Since multiplication by s in the Laplace domain corresponds to $\partial/\partial t$, (C.3) inverts to the visco-elastic Stokes kernel (27).

APPENDIX D: NUMERICAL EVALUATION OF THE EVE INTEGRALS

The EVE step integrals (33) and (34) are evaluated with adaptive Gauss–Kronrod quadrature (Laurie 1997), with integration breakpoints at the successive zeros of $\sin(kx)$, $k = n\pi/x$ ($n = 1, 2, \ldots$), which partition the oscillatory integrand into well-behaved sub-intervals. The upper truncation wavenumber is set adaptively so that the exponential factor is negligible at the cutoff.

The EVE RSS integrals (40) and (41) contain factors that tend to finite, non-zero limits as $k \to \infty$, so their integrands decay slowly. For the fits of Section 4, the large-k limit of each integrand is subtracted and integrated in closed form with the Dirichlet integral, which also gives the axis jump (43). The remainder decays as $1/k^2$ on the scale $k \sim 1/(a\tau_R)$ and is integrated with the same segmented quadrature up to the fixed cutoff $10^4/(a\tau_R)$.

APPENDIX E: STOCHASTIC RECURRENCE EXPERIMENTS

Each realisation of Section 5.5 sums the EV step velocities (15) of 300 past intrusions and is evaluated at phase $\tau/T = 0.5$; each case comprises 500 realisations. Intervals T_j and openings b_j follow Normal distributions with coefficient of variation $c_v = 0.30$, truncated at 5 per cent of the mean. Dike positions x_j follow a Normal distribution of standard deviation $\sigma_x = 0.1\,aT$; a uniform distribution with the same standard deviation gives 10–90 per cent bands within 10 per cent of the Normal case.